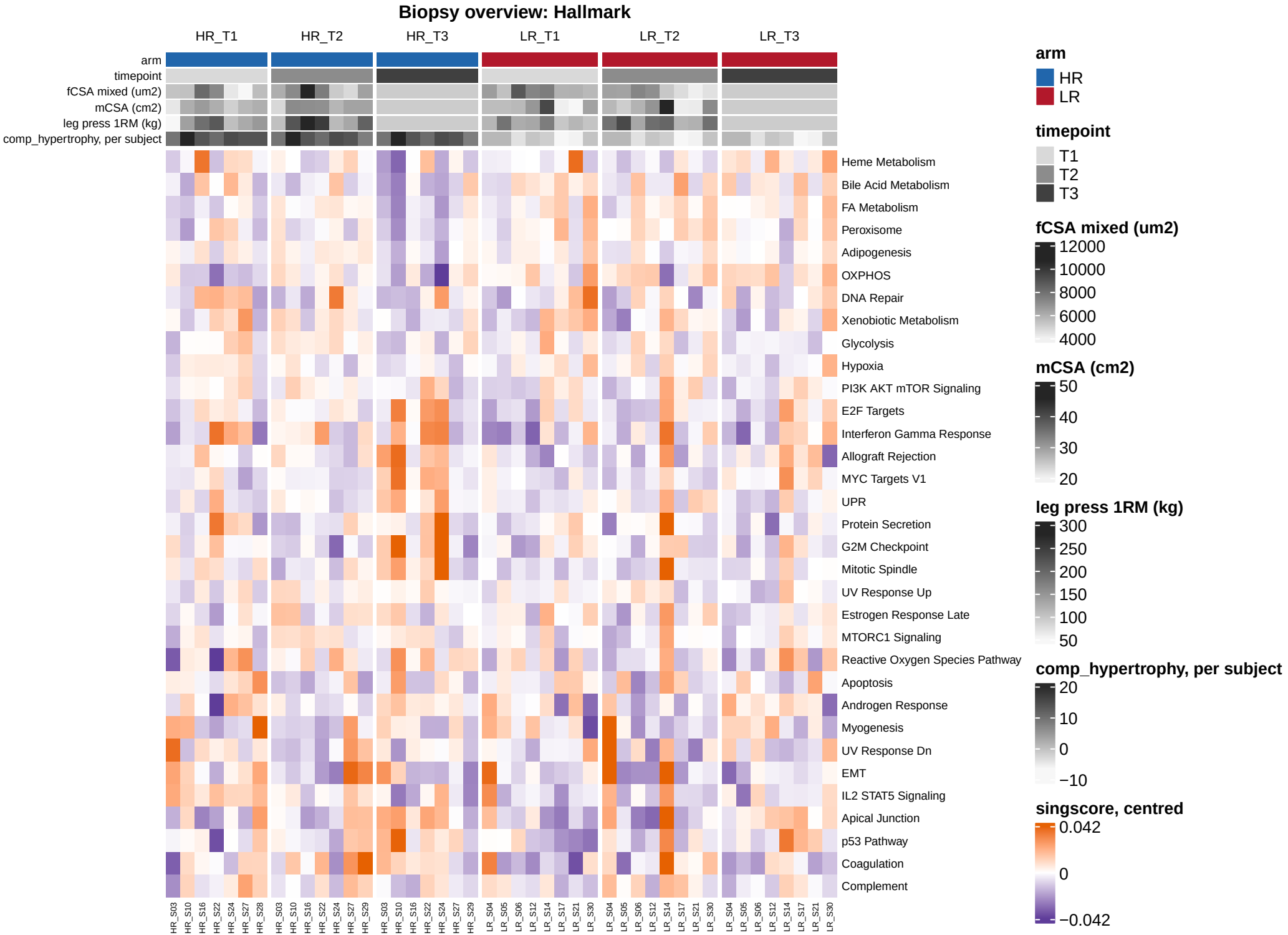
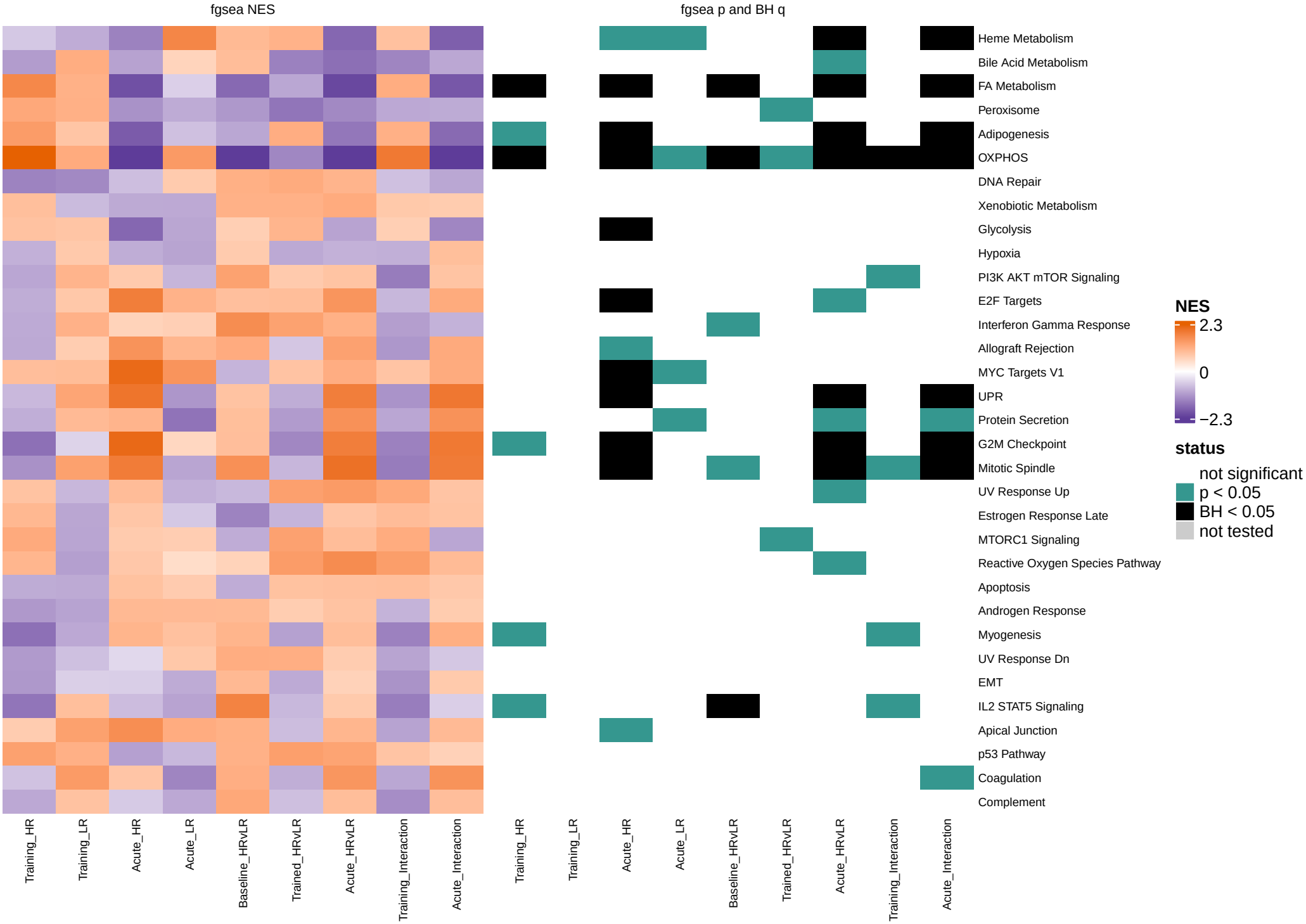
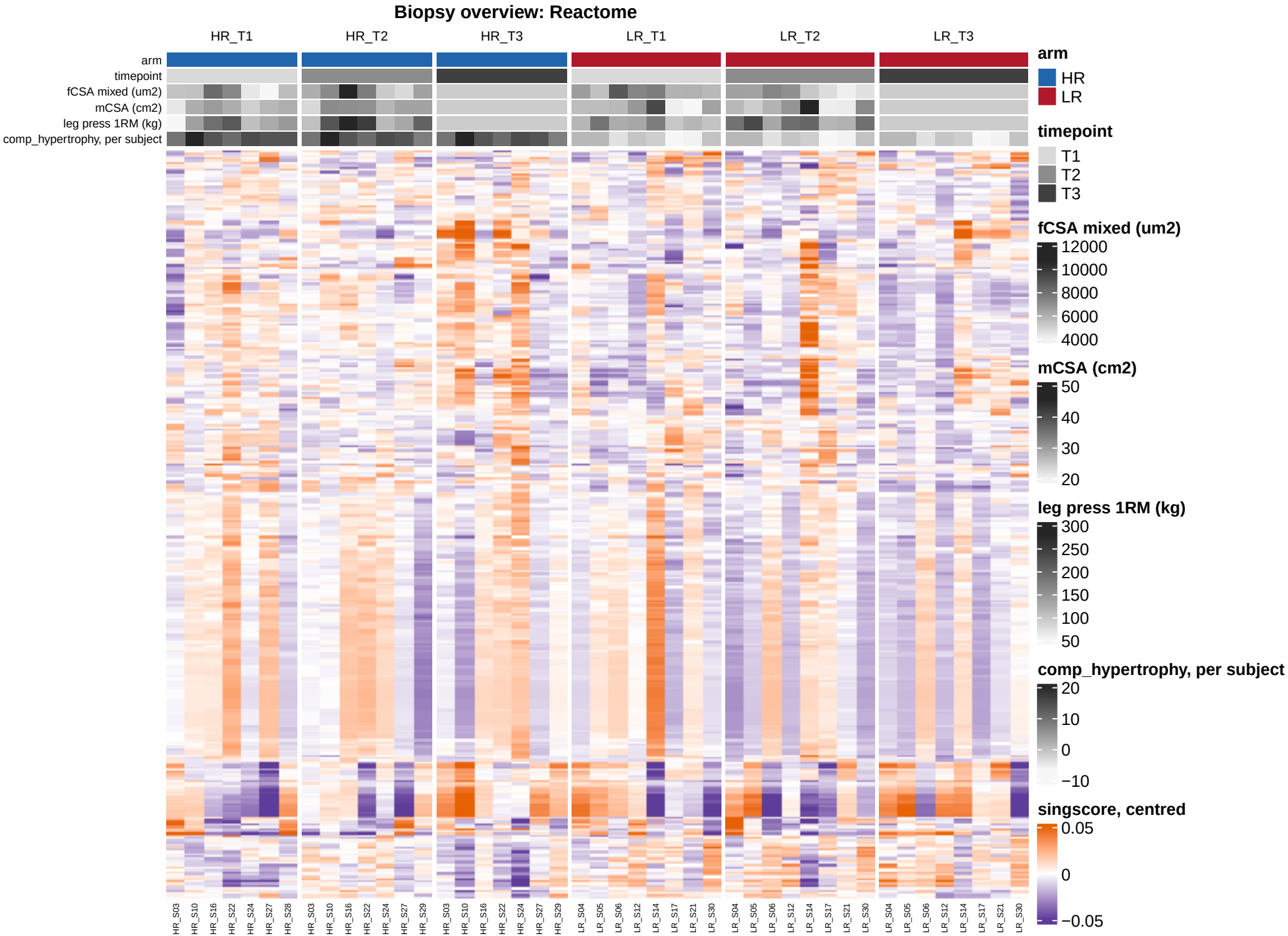



All 33 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.042. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

# Contrast overview: Hallmark

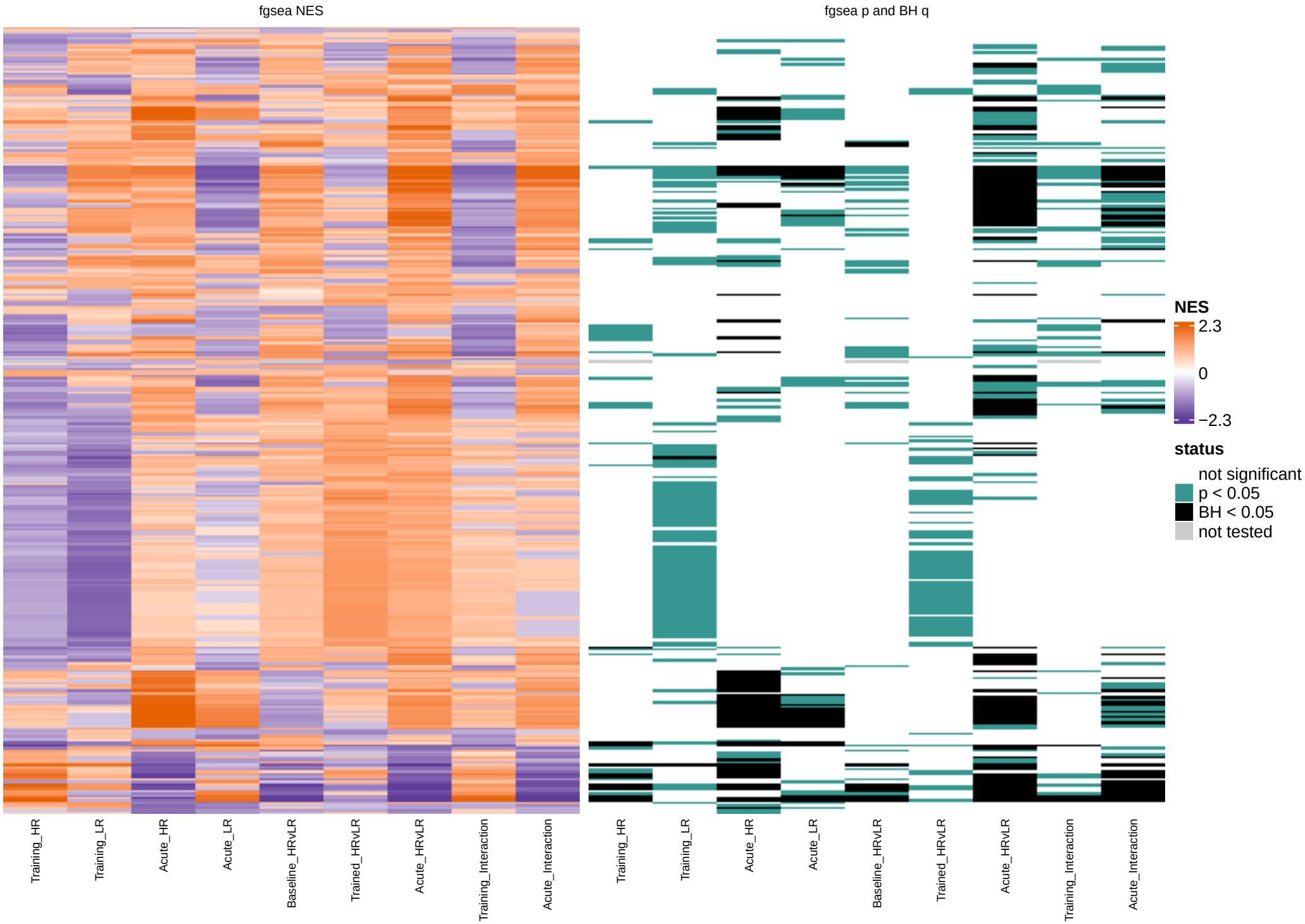



All 33 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.3. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

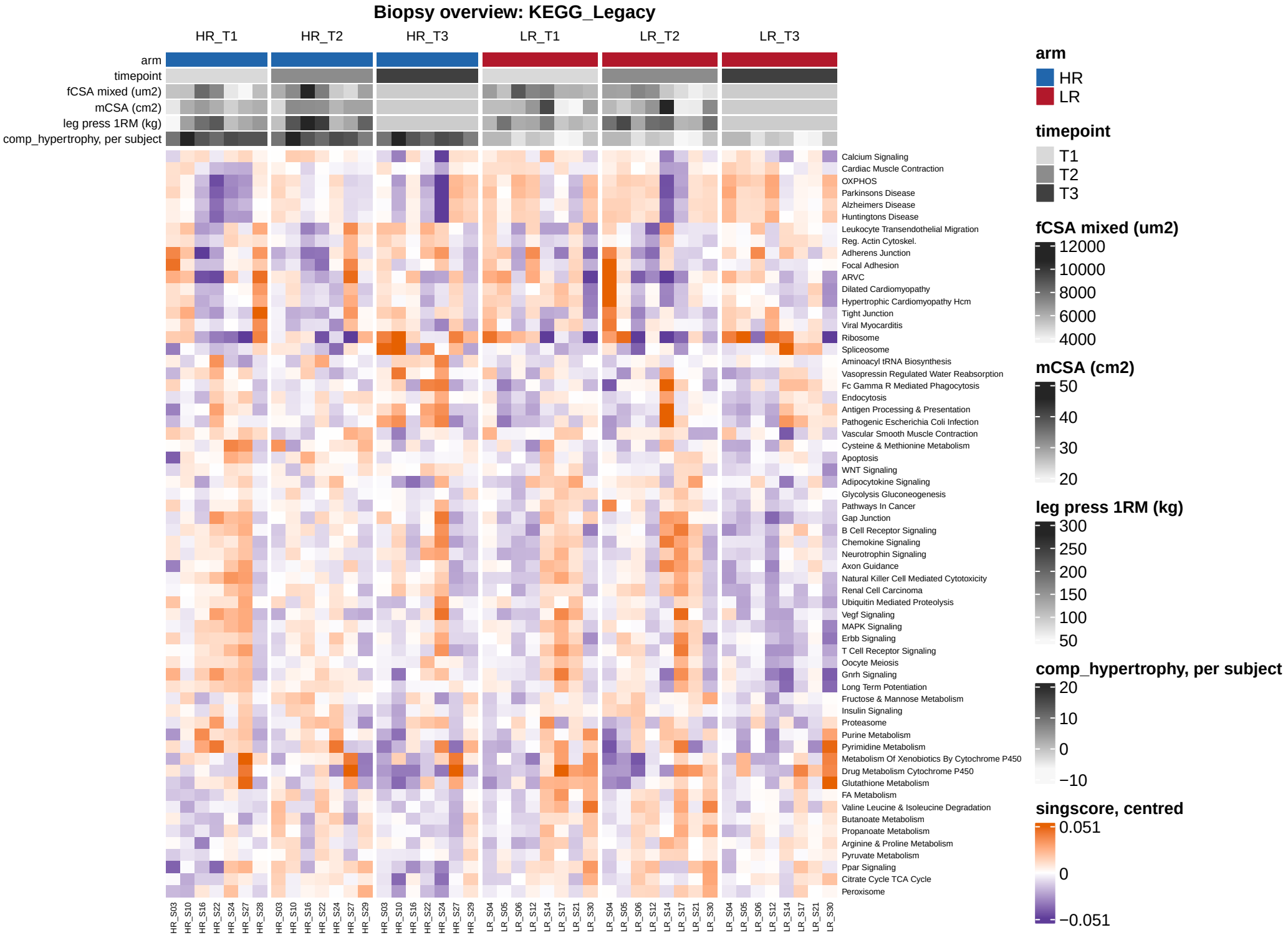



All 328 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.05. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

# Contrast overview: Reactome

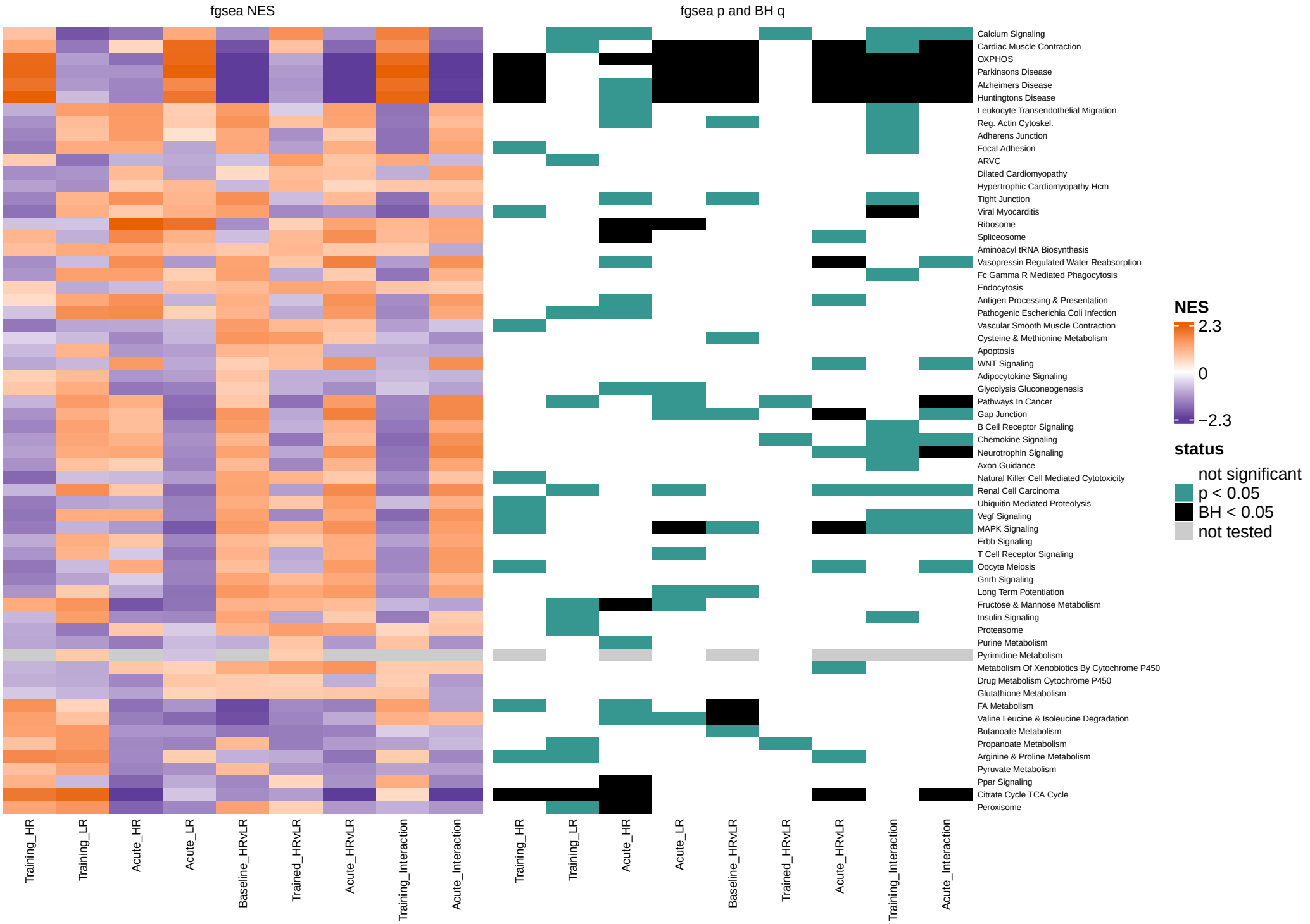



All 328 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.3. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

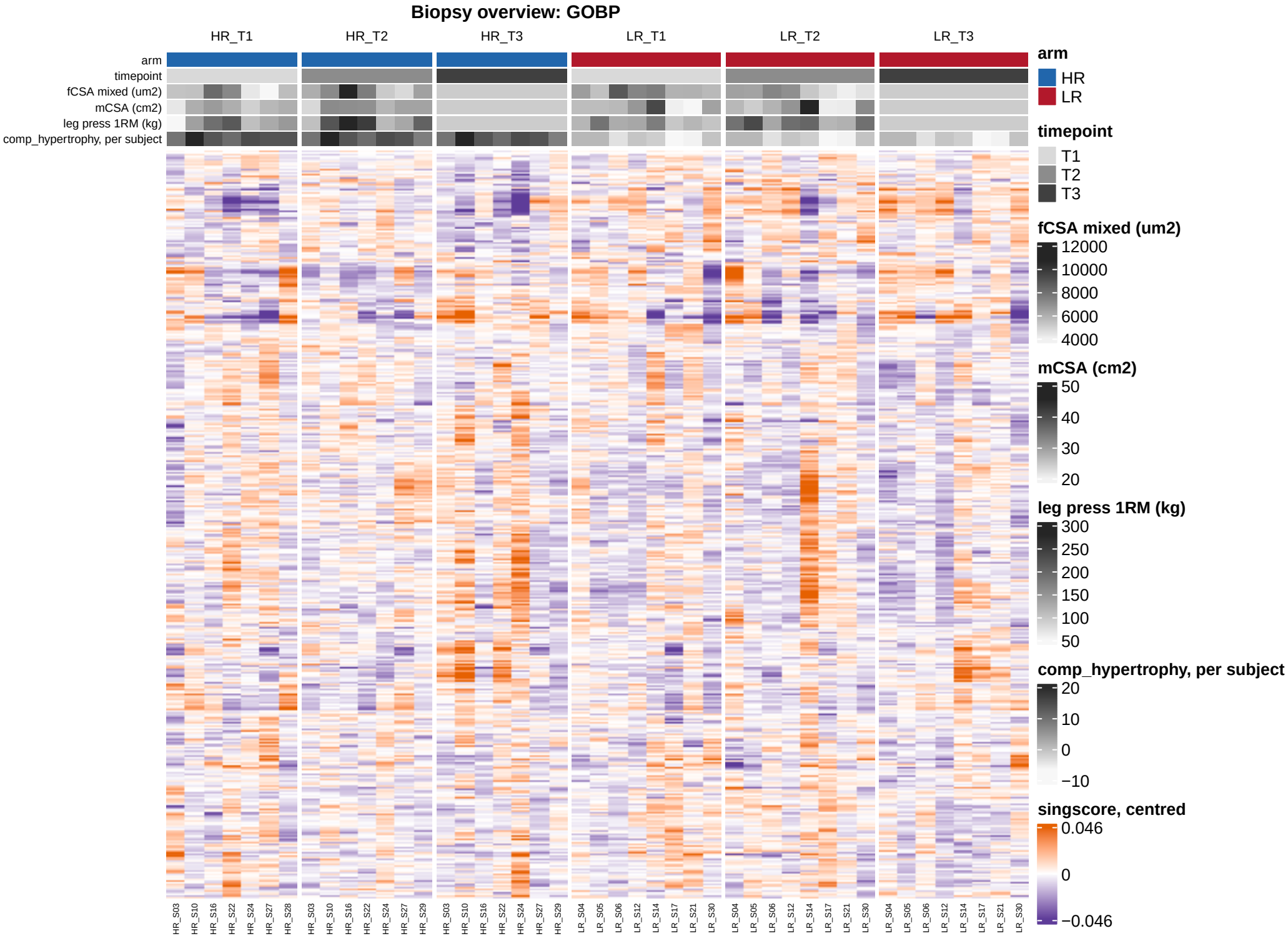



All 62 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.051. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

# Contrast overview: KEGG_Legacy

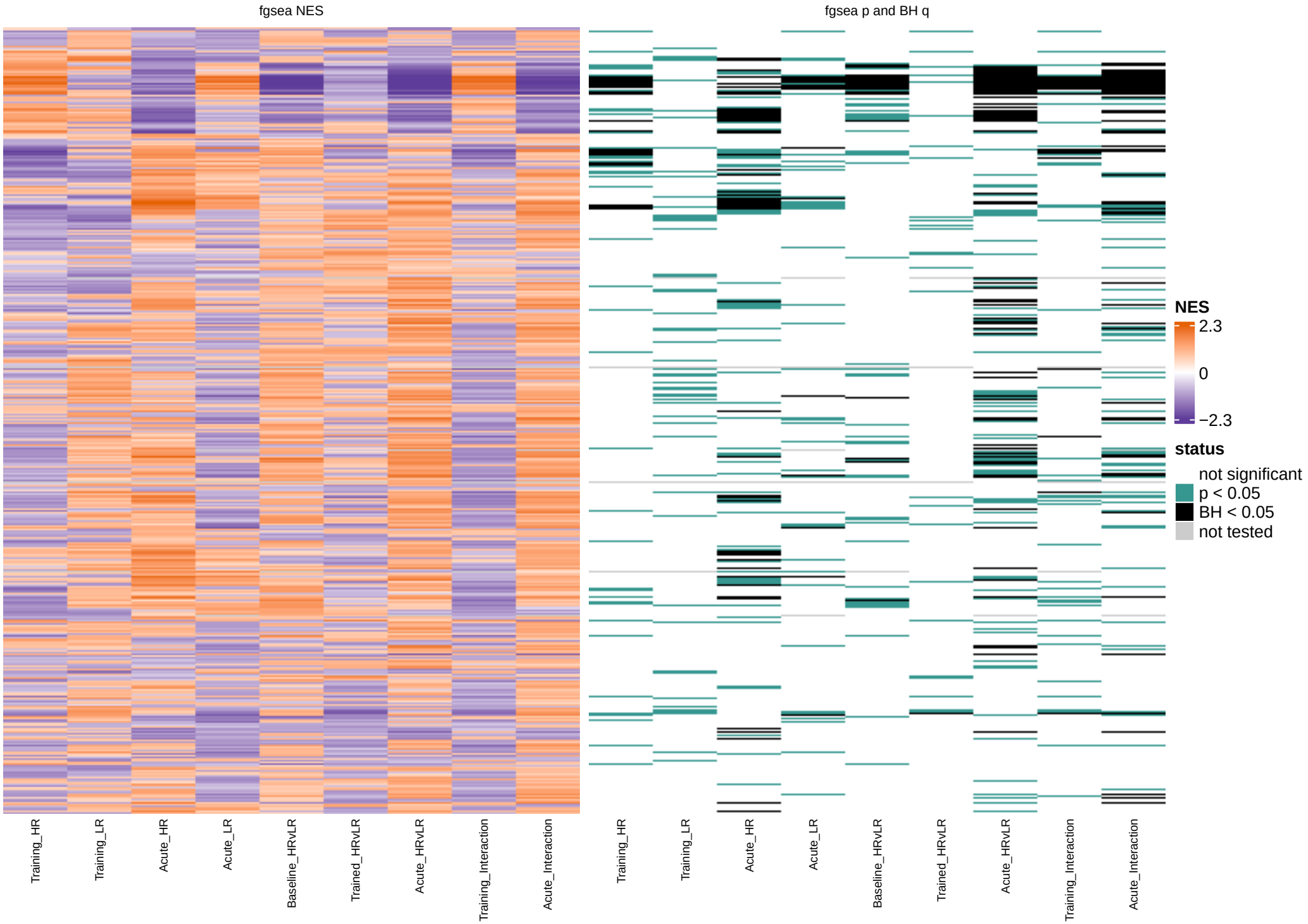



All 62 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.3. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.



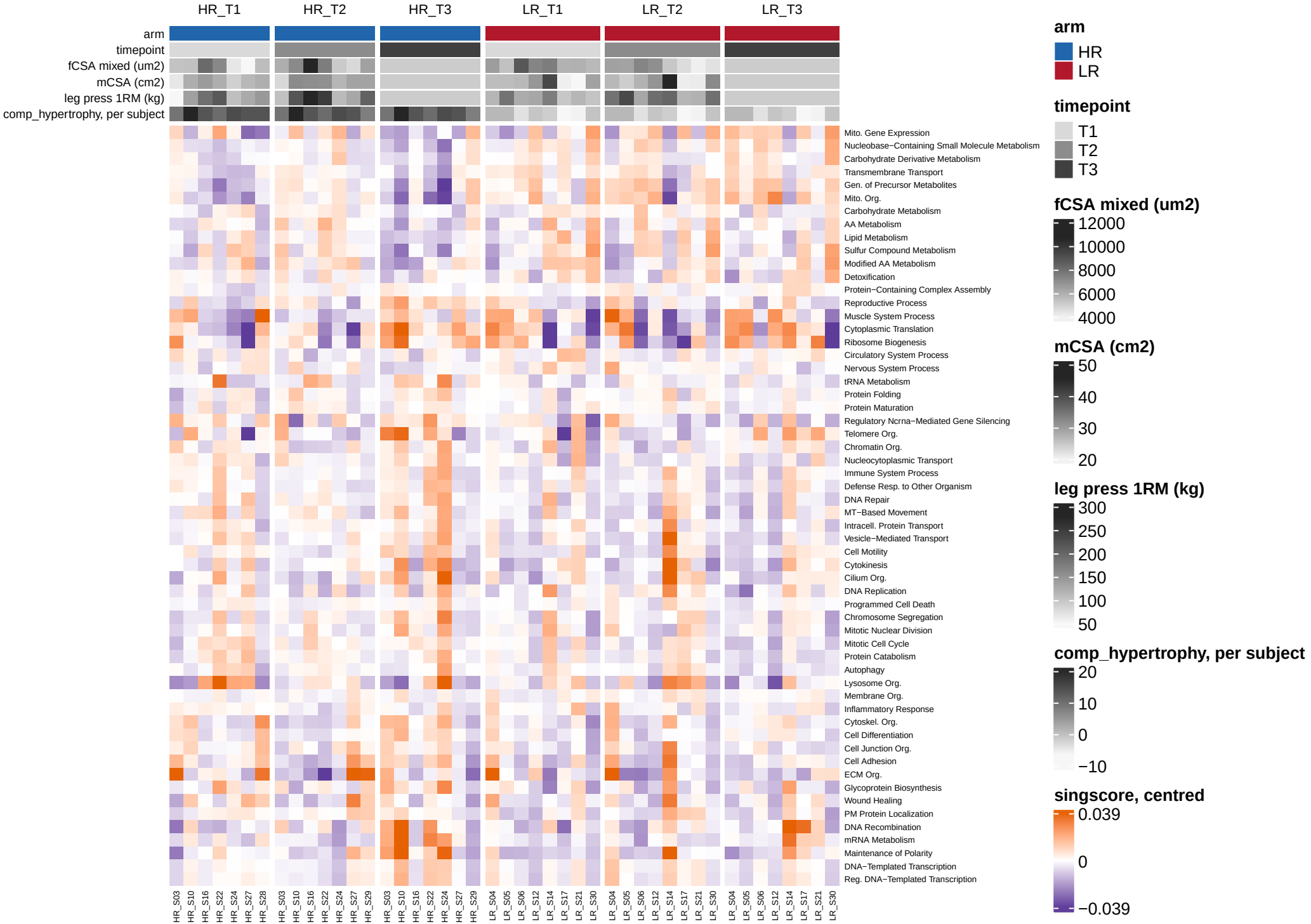

All 897 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.046. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

# Contrast overview: GOBP



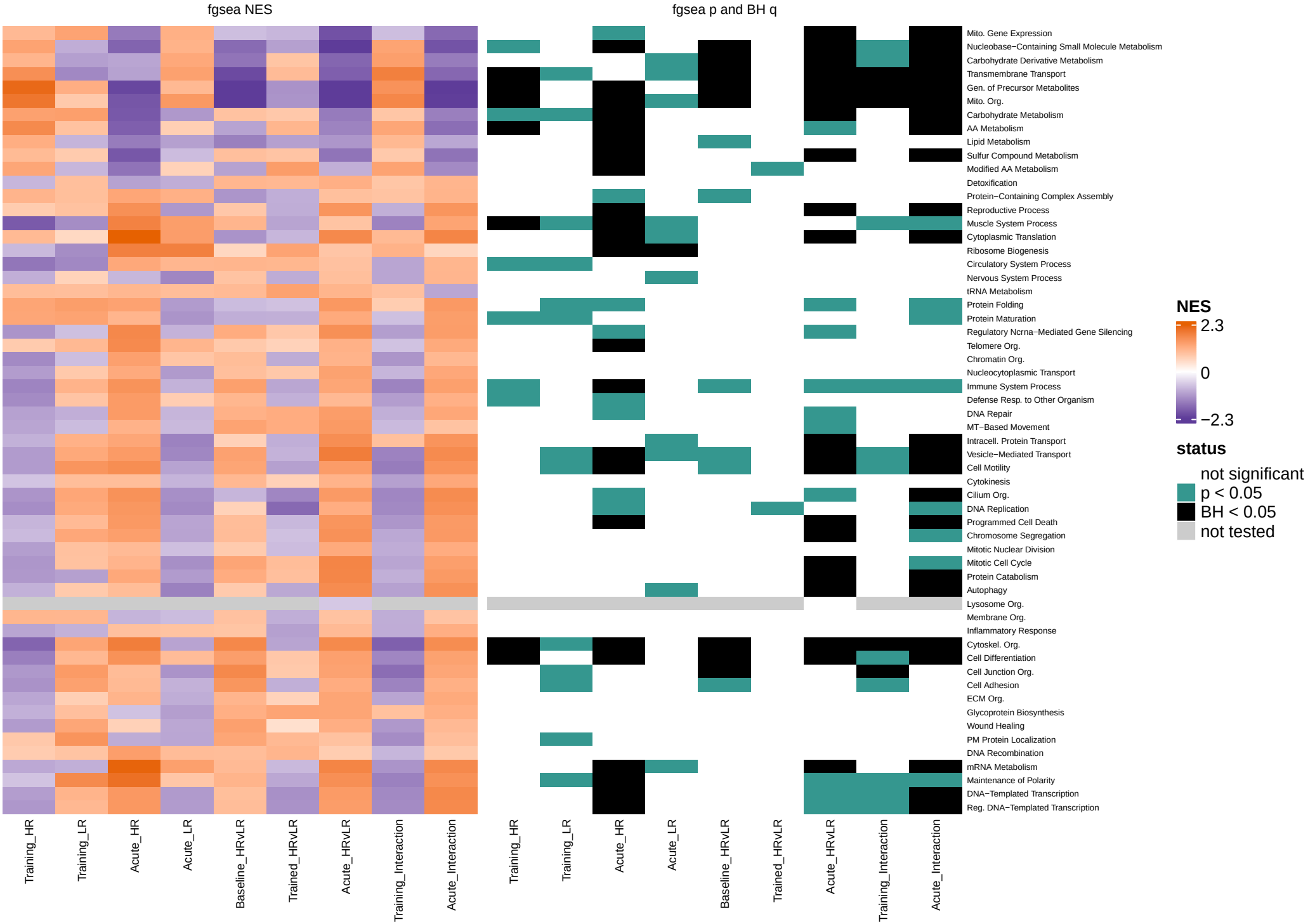

All 897 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.3. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

# Biopsy overview: GO_Slim

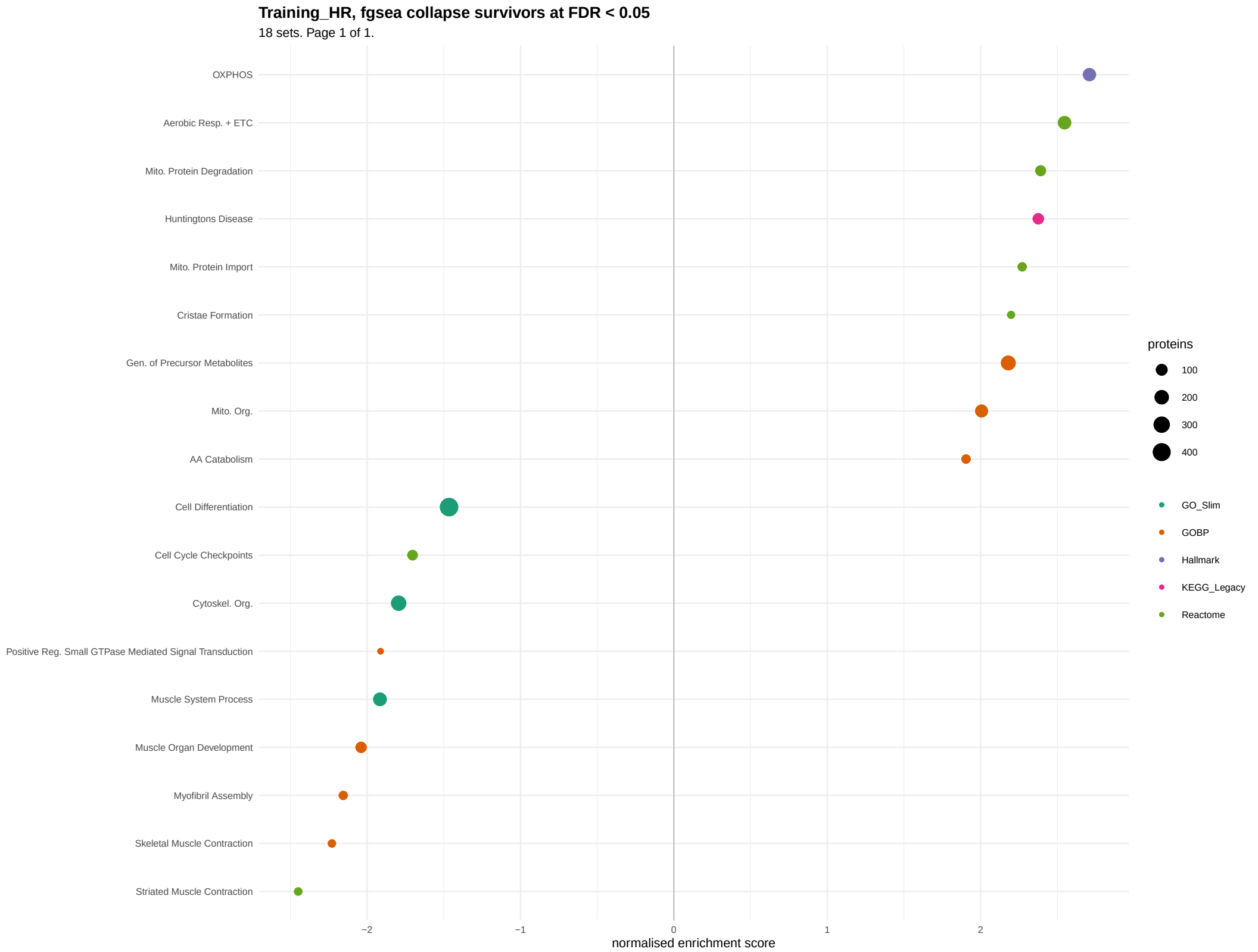



All 58 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.039. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

# Contrast overview: GO_Slim

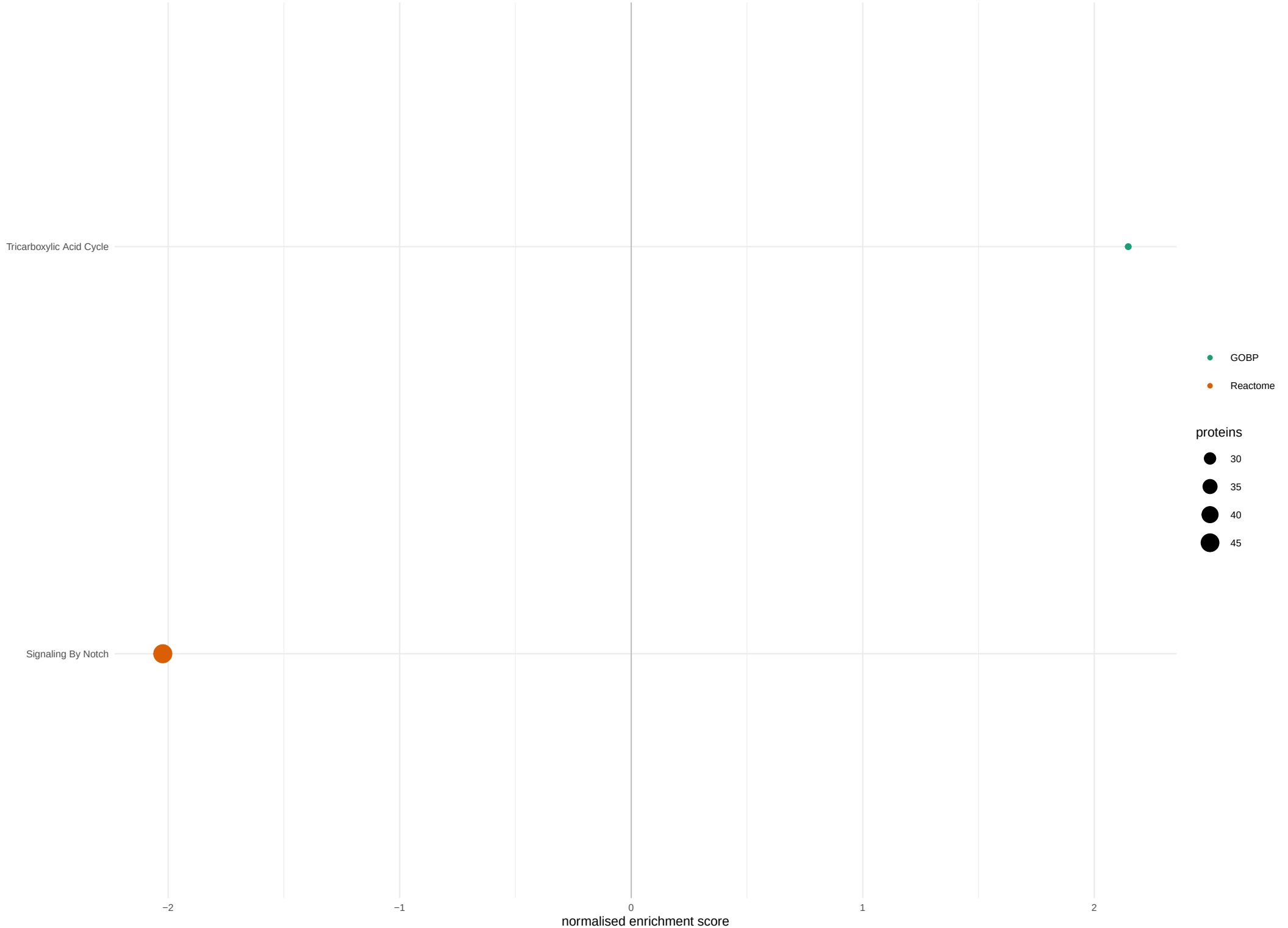



All 58 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.3. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.


Training_HR, fgsea collapse survivors at FDR < 0.05
18 sets. Page 1 of 1.
OXPHOS
Aerobic Resp. + ETC
Mito. Protein Degradation
Huntingtons Disease
Mito. Protein Import
Cristae Formation
Gen. of Precursor Metabolites
Mito. Org.
AA Catabolism
Cell Differentiation
Cell Cycle Checkpoints
Cytoskel. Org.
Positive Reg. Small GTPase Mediated Signal Transduction
Muscle System Process
Muscle Organ Development
Myofibril Assembly
Skeletal Muscle Contraction
Striated Muscle Contraction
−2
−1
0
1
2
normalised enrichment score
proteins
100
200
300
400
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

**Training_LR, fgsea collapse survivors at FDR < 0.05**

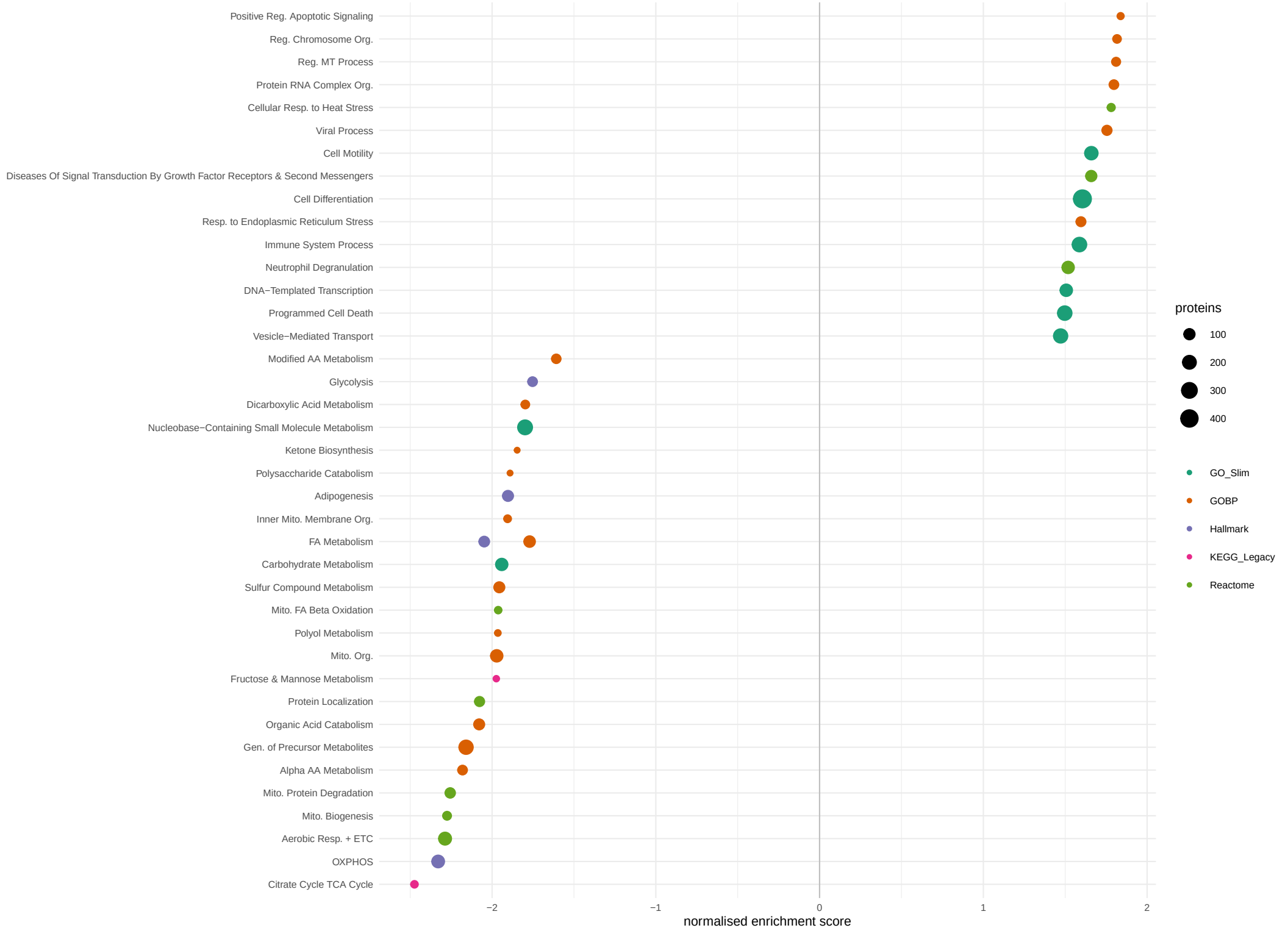
2 sets. Page 1 of 1.



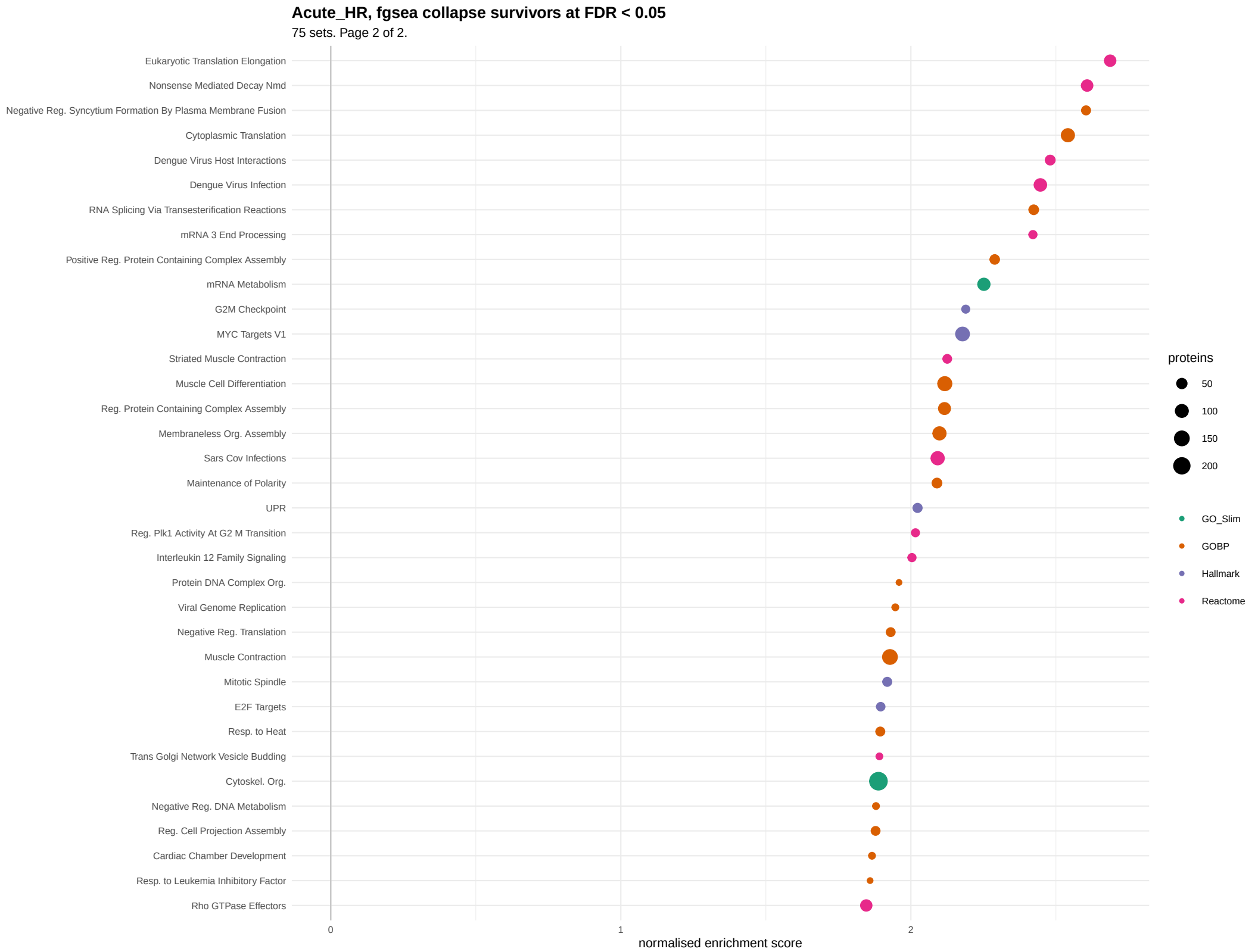

fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

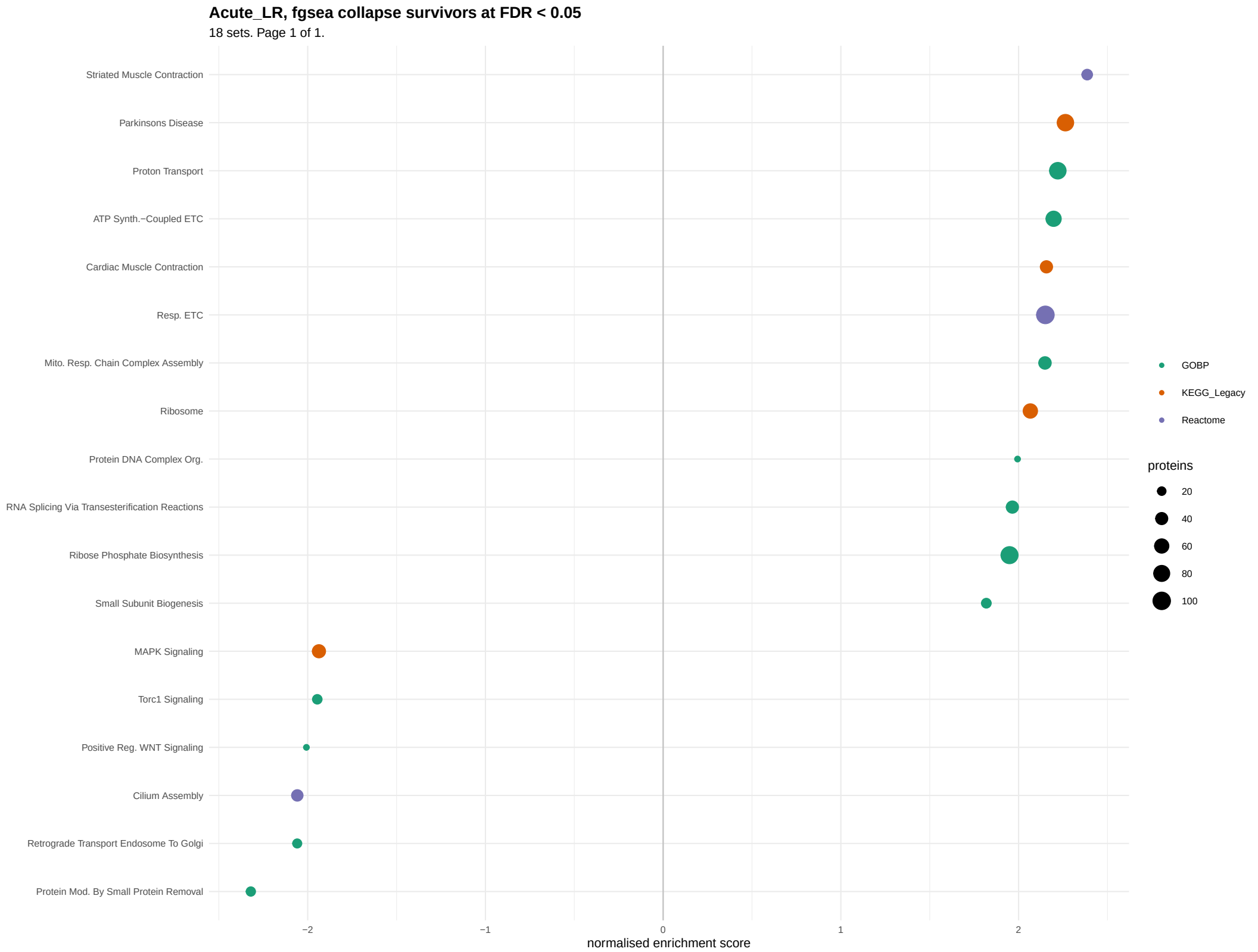
# Acute_HR, fgsea collapse survivors at FDR < 0.05

75 sets. Page 1 of 2.



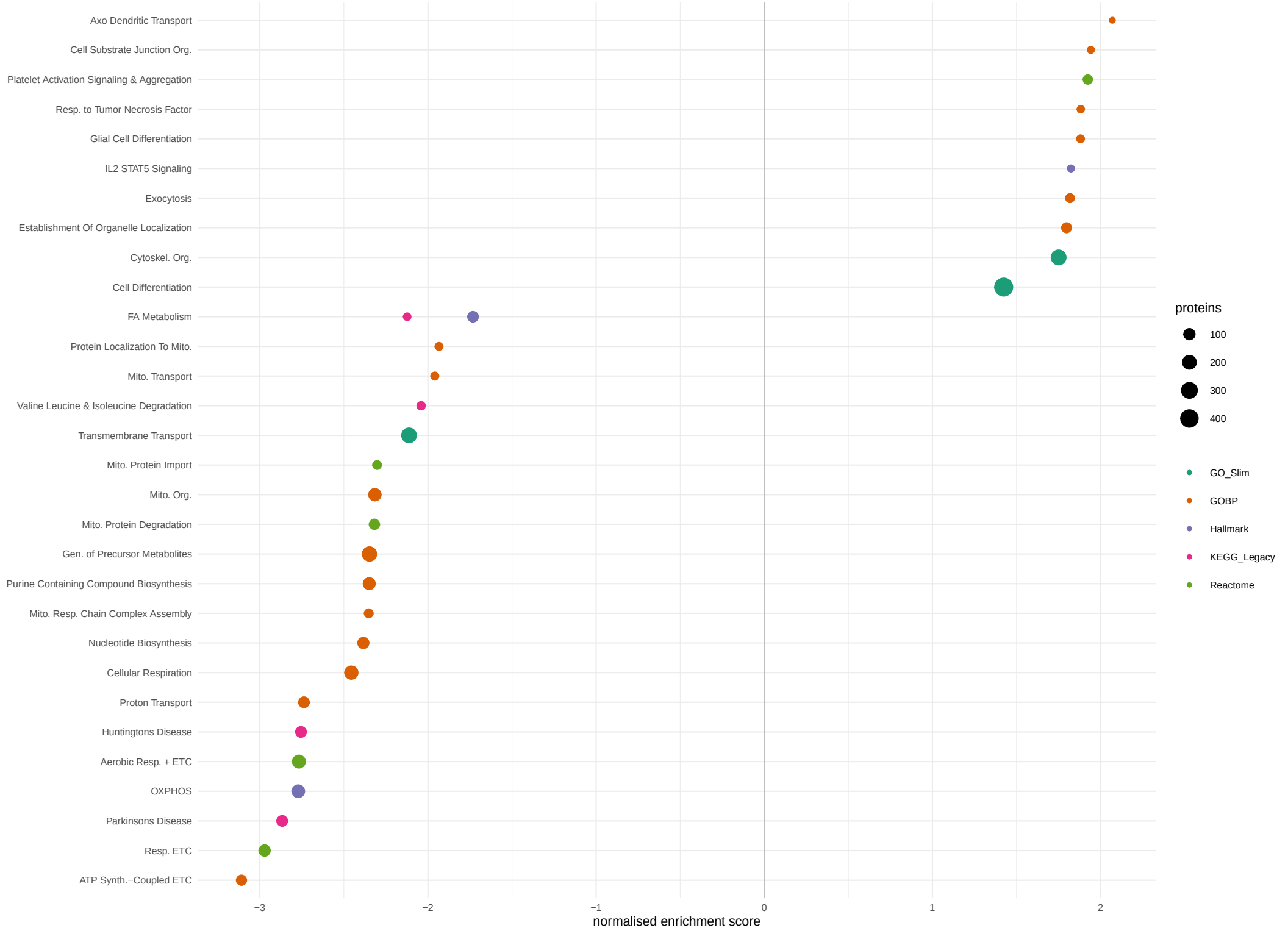

fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.


Acute_HR, fgsea collapse survivors at FDR < 0.05
75 sets. Page 2 of 2.
Eukaryotic Translation Elongation
Nonsense Mediated Decay Nmd
Negative Reg. Syncytium Formation By Plasma Membrane Fusion
Cytoplasmic Translation
Dengue Virus Host Interactions
Dengue Virus Infection
RNA Splicing Via Transesterification Reactions
mRNA 3 End Processing
Positive Reg. Protein Containing Complex Assembly
mRNA Metabolism
G2M Checkpoint
MYC Targets V1
Striated Muscle Contraction
Muscle Cell Differentiation
Reg. Protein Containing Complex Assembly
Membraneless Org. Assembly
Sars Cov Infections
Maintenance of Polarity
UPR
Reg. Plk1 Activity At G2 M Transition
Interleukin 12 Family Signaling
Protein DNA Complex Org.
Viral Genome Replication
Negative Reg. Translation
Muscle Contraction
Mitotic Spindle
E2F Targets
Resp. to Heat
Trans Golgi Network Vesicle Budding
Cytoskel. Org.
Negative Reg. DNA Metabolism
Reg. Cell Projection Assembly
Cardiac Chamber Development
Resp. to Leukemia Inhibitory Factor
Rho GTPase Effectors
0
1
2
normalised enrichment score
proteins
50
100
150
200
GO_Slim
GOBP
Hallmark
Reactome
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.


Acute_LR, fgsea collapse survivors at FDR < 0.05
18 sets. Page 1 of 1.
Striated Muscle Contraction
Parkinsons Disease
Proton Transport
ATP Synth.−Coupled ETC
Cardiac Muscle Contraction
Resp. ETC
Mito. Resp. Chain Complex Assembly
Ribosome
Protein DNA Complex Org.
RNA Splicing Via Transesterification Reactions
Ribose Phosphate Biosynthesis
Small Subunit Biogenesis
MAPK Signaling
Torc1 Signaling
Positive Reg. WNT Signaling
Cilium Assembly
Retrograde Transport Endosome To Golgi
Protein Mod. By Small Protein Removal
−2
−1
0
1
2
normalised enrichment score
GOBP
KEGG_Legacy
Reactome
proteins
20
40
60
80
100
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

**Baseline_HRvLR, fgsea collapse survivors at FDR < 0.05**

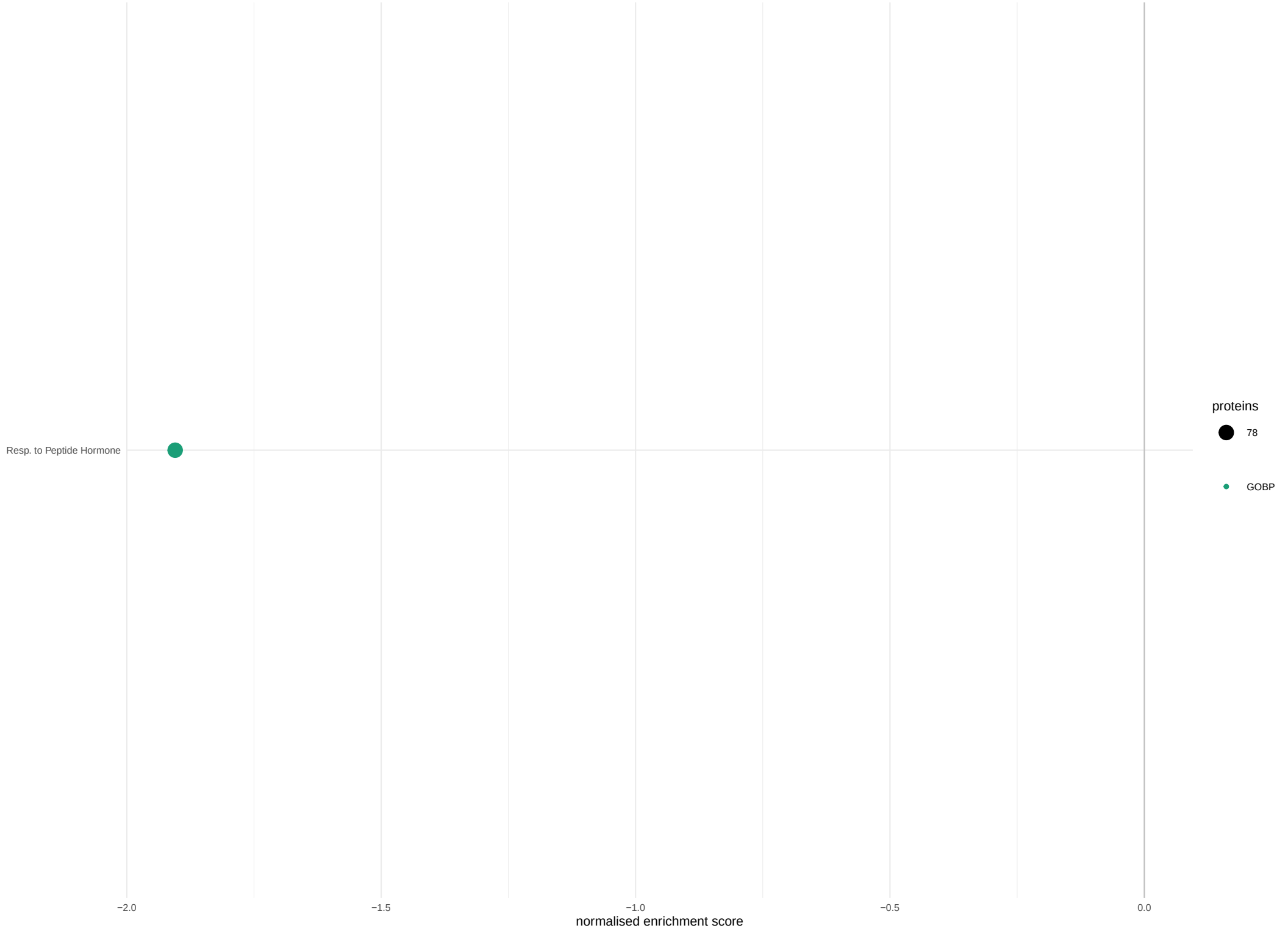
31 sets. Page 1 of 1.




fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

**Trained_HRvLR, fgsea collapse survivors at FDR < 0.05**

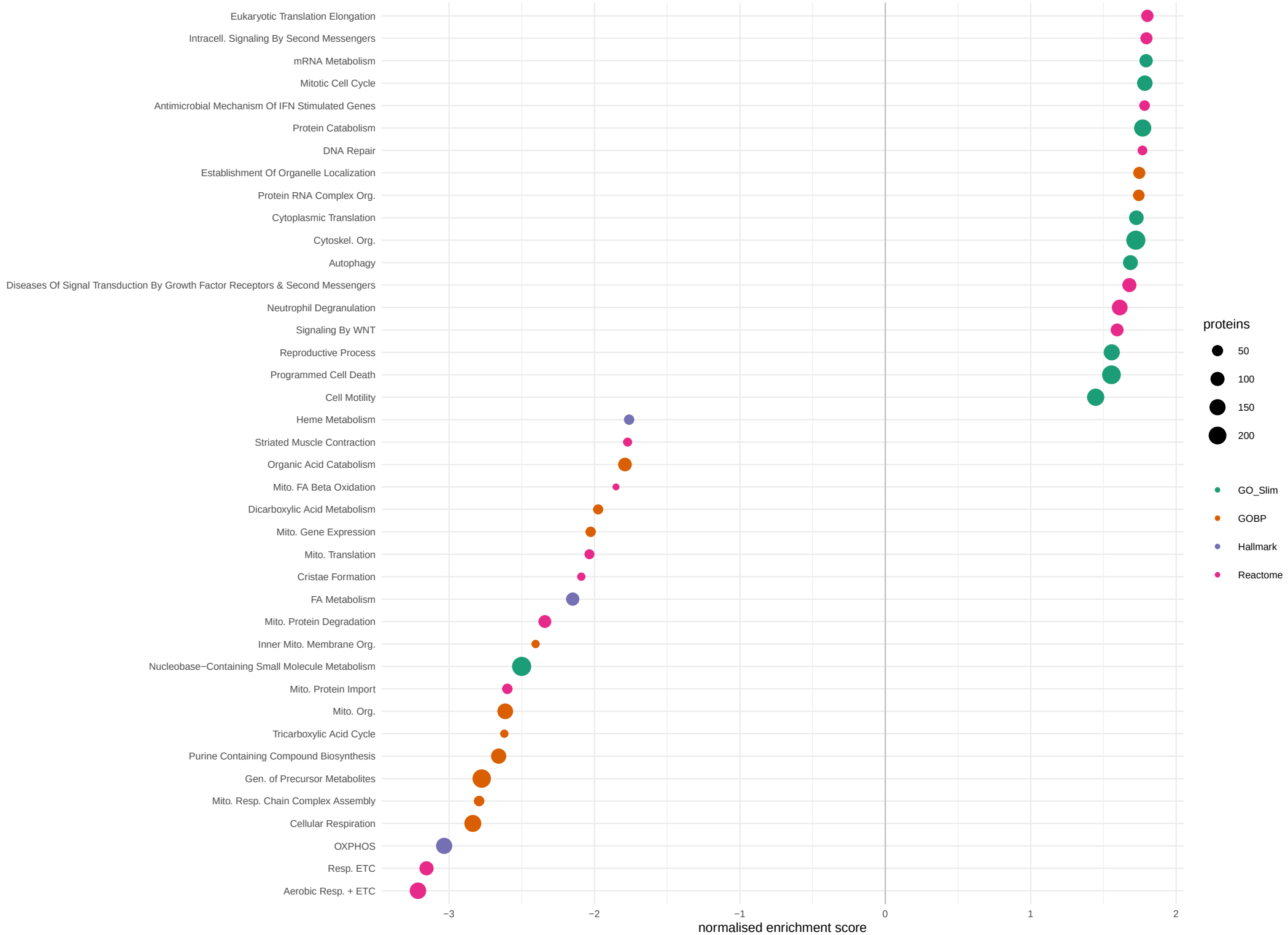
1 sets. Page 1 of 1.




fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

# Acute_HRvLR, fgsea collapse survivors at FDR < 0.05

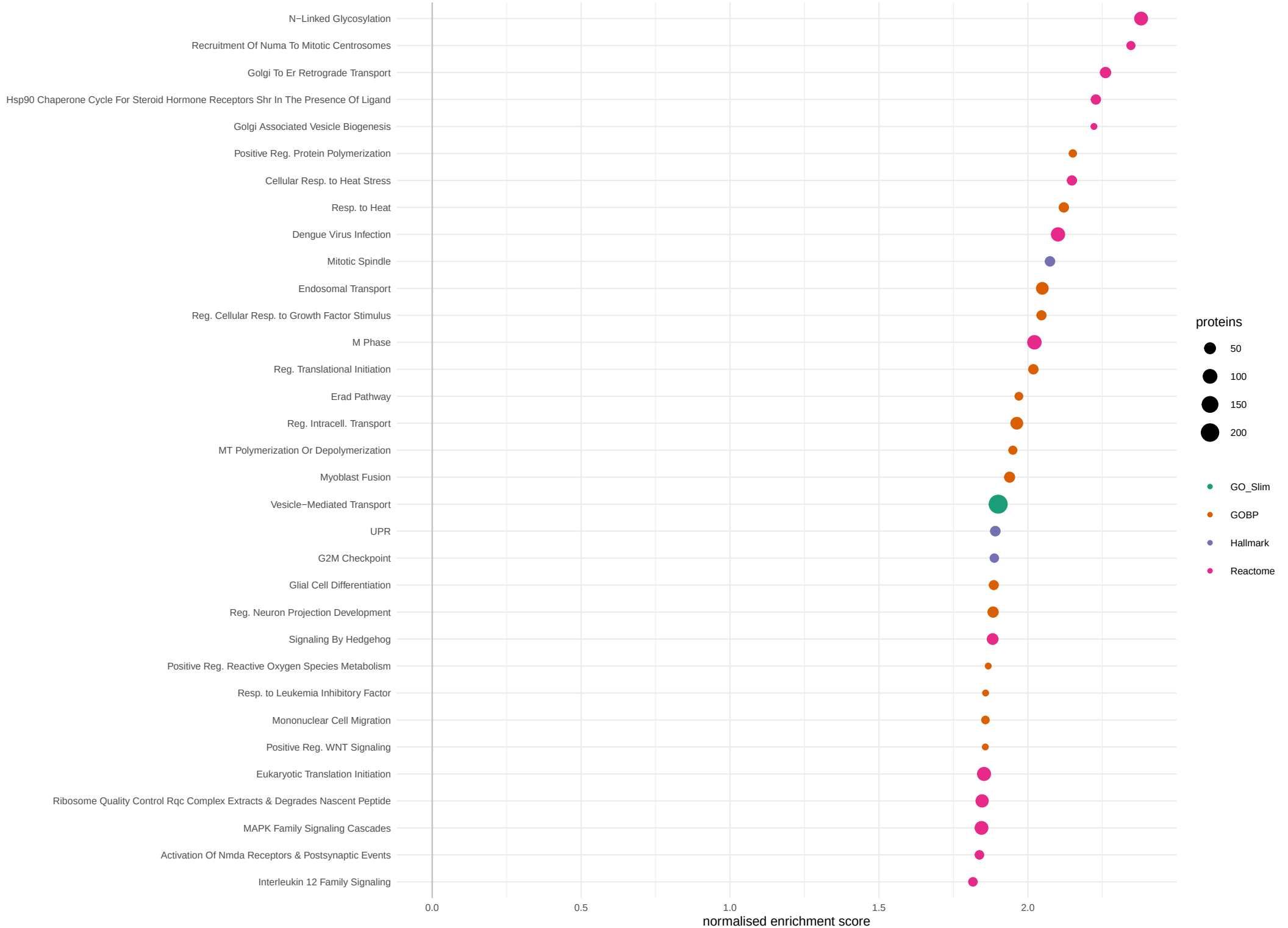
73 sets. Page 1 of 2.




fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

# Acute_HRvLR, fgsea collapse survivors at FDR < 0.05

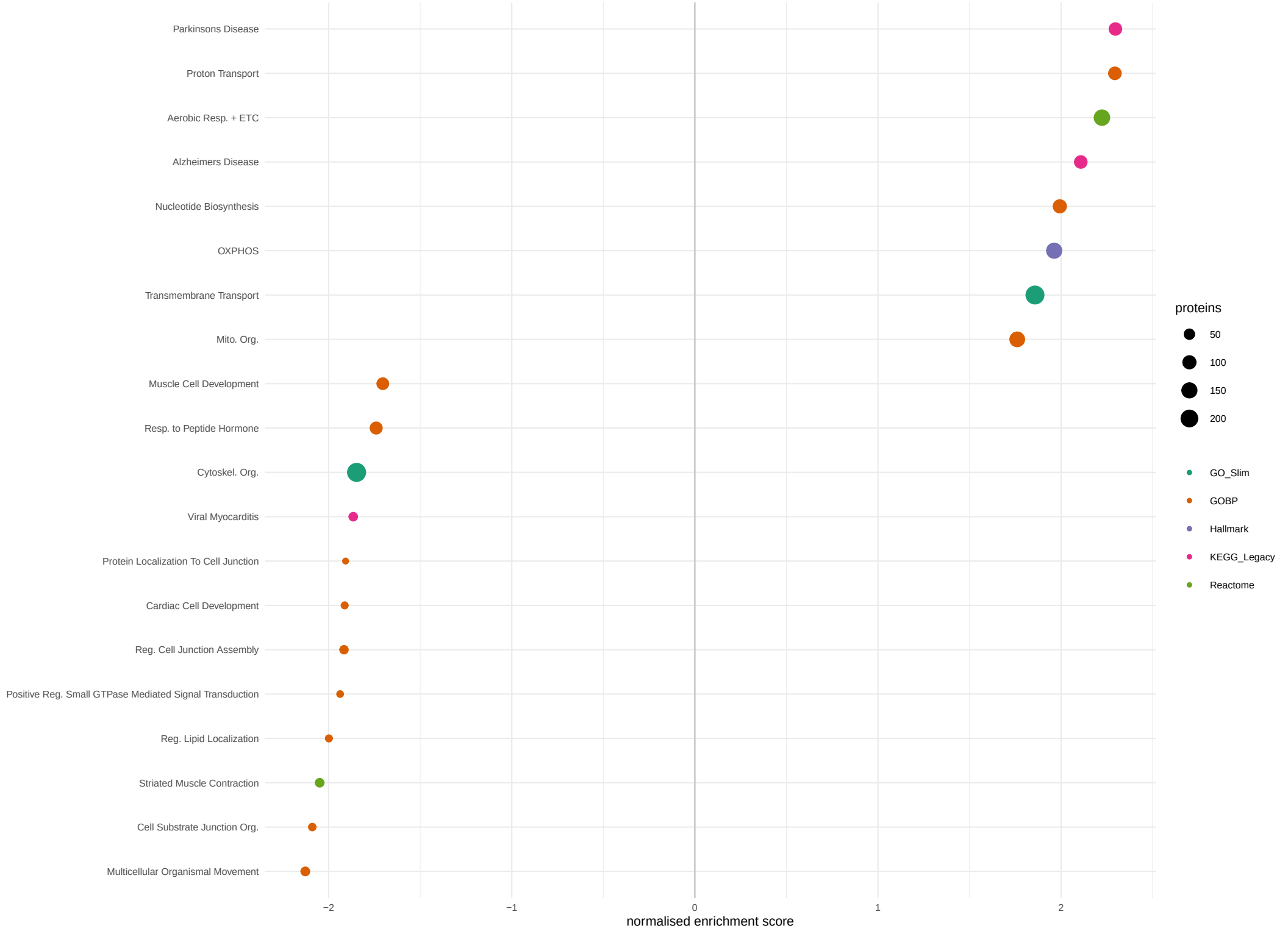
73 sets. Page 2 of 2.



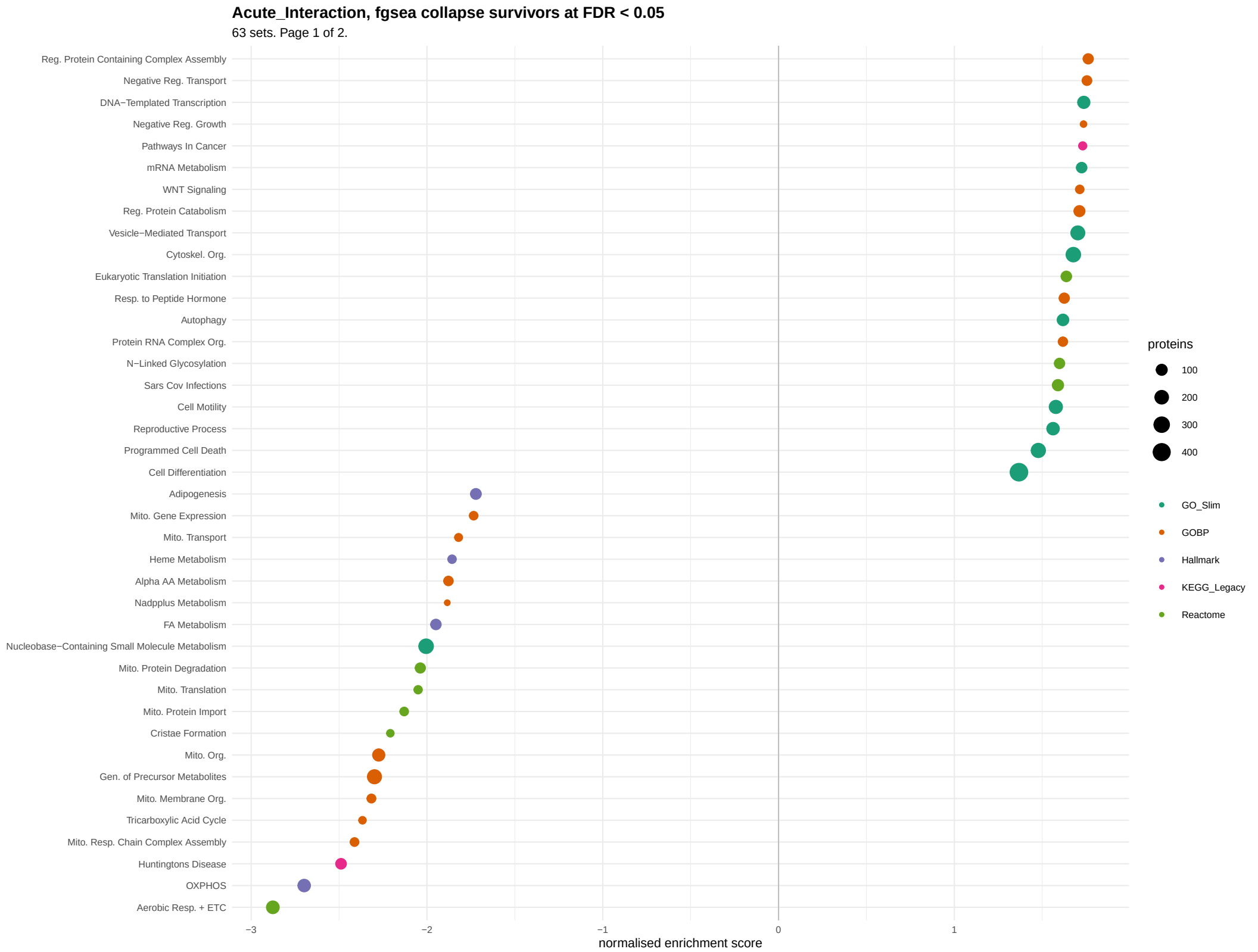

fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

**Training_Interaction, fgsea collapse survivors at FDR < 0.05**

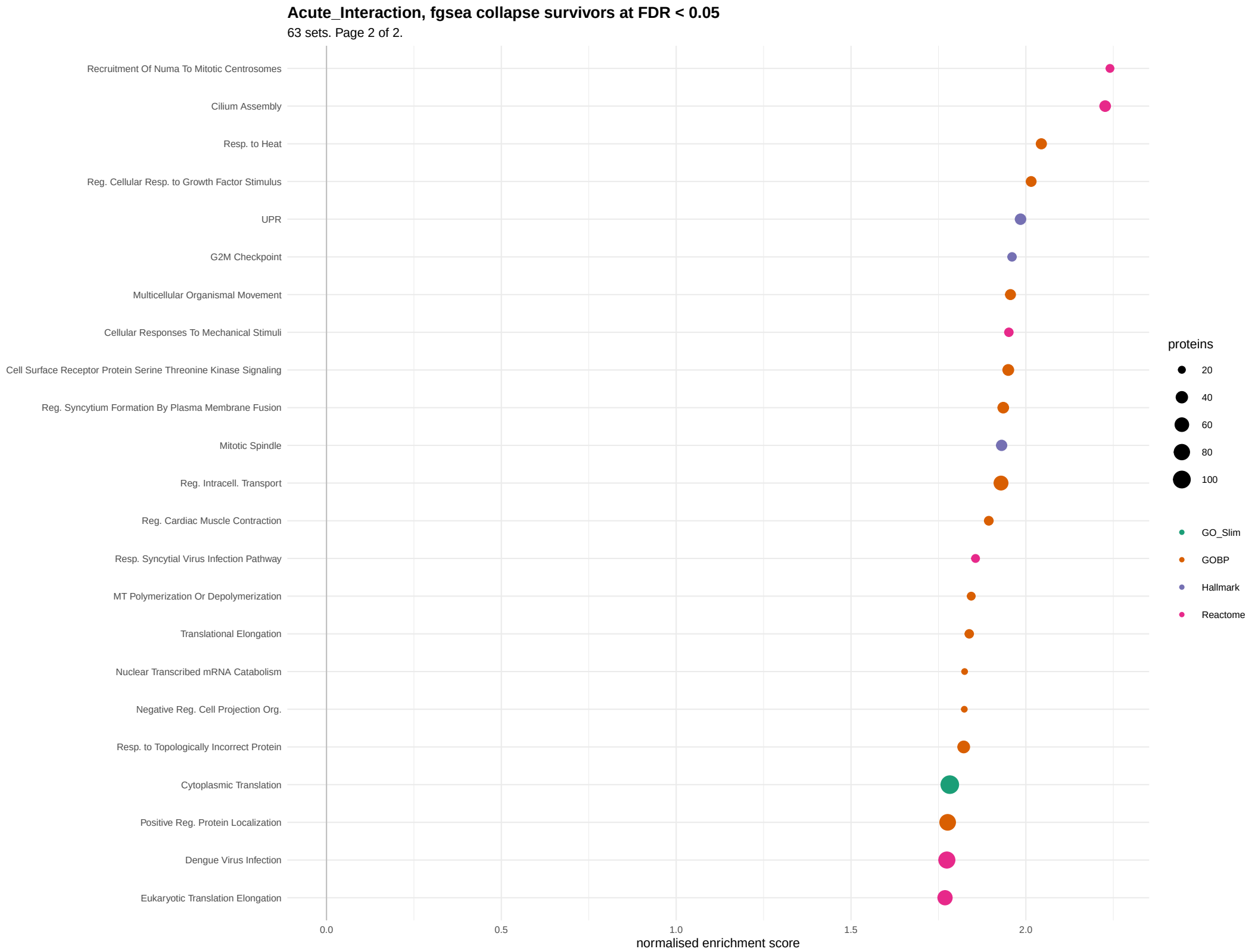
20 sets. Page 1 of 1.

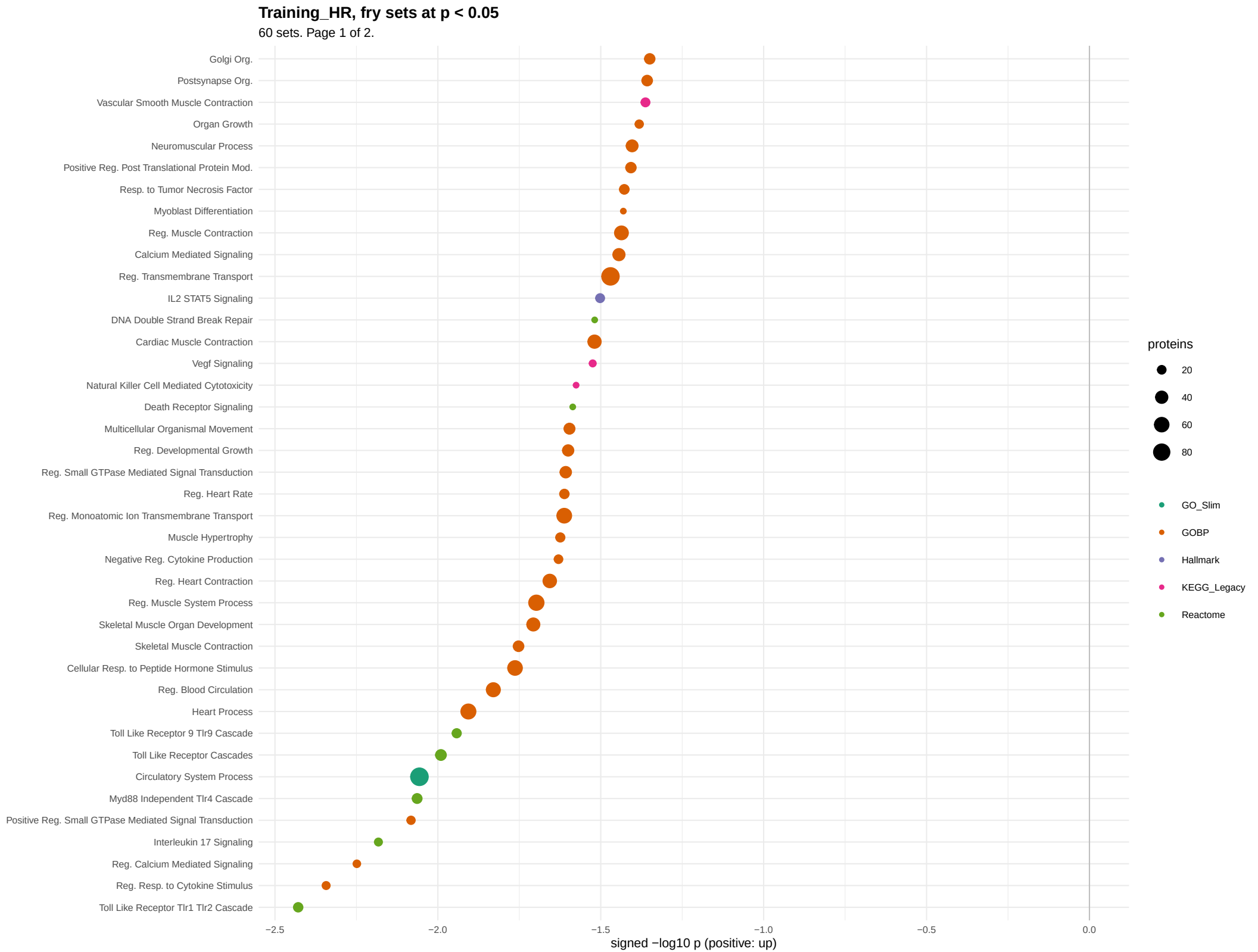



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

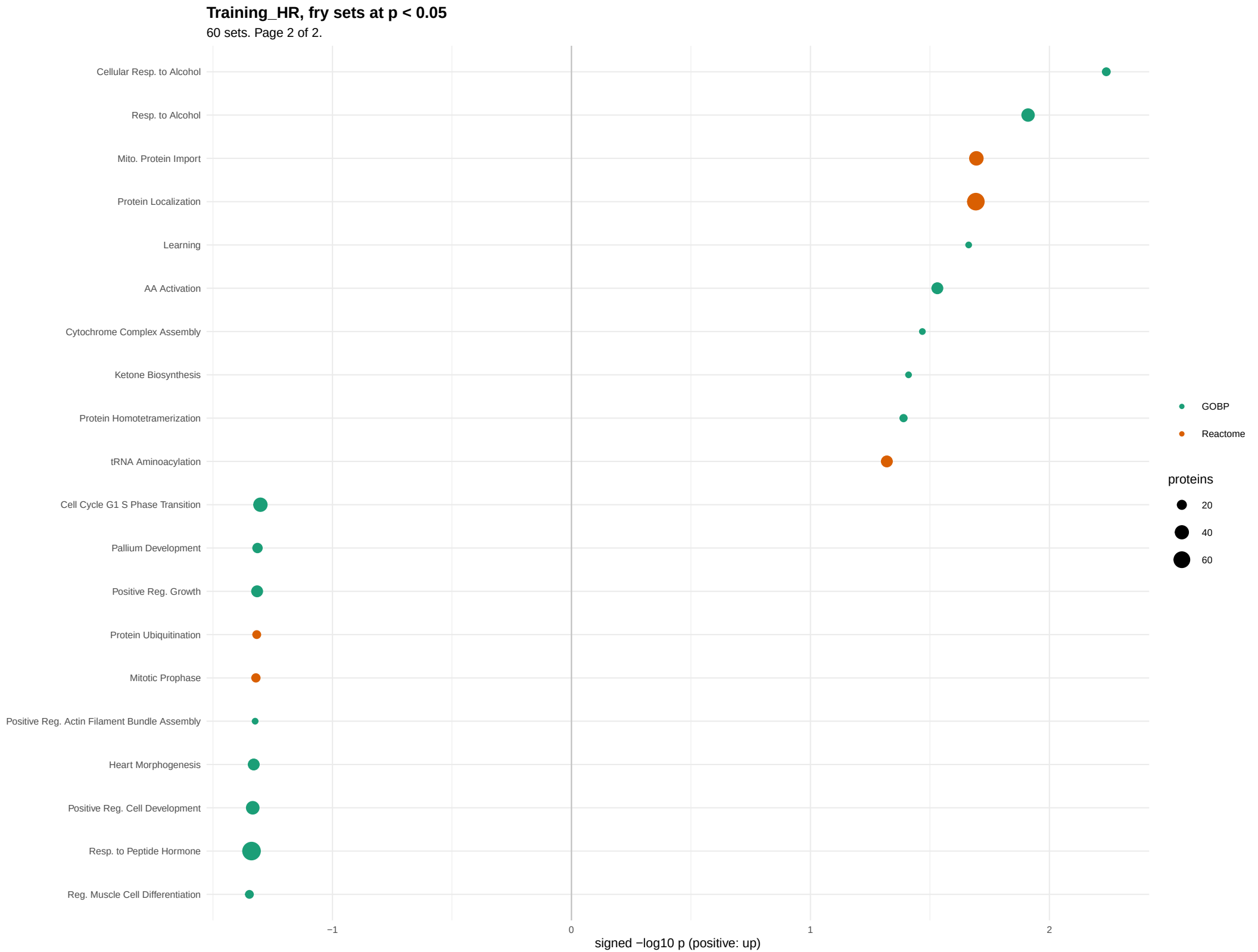


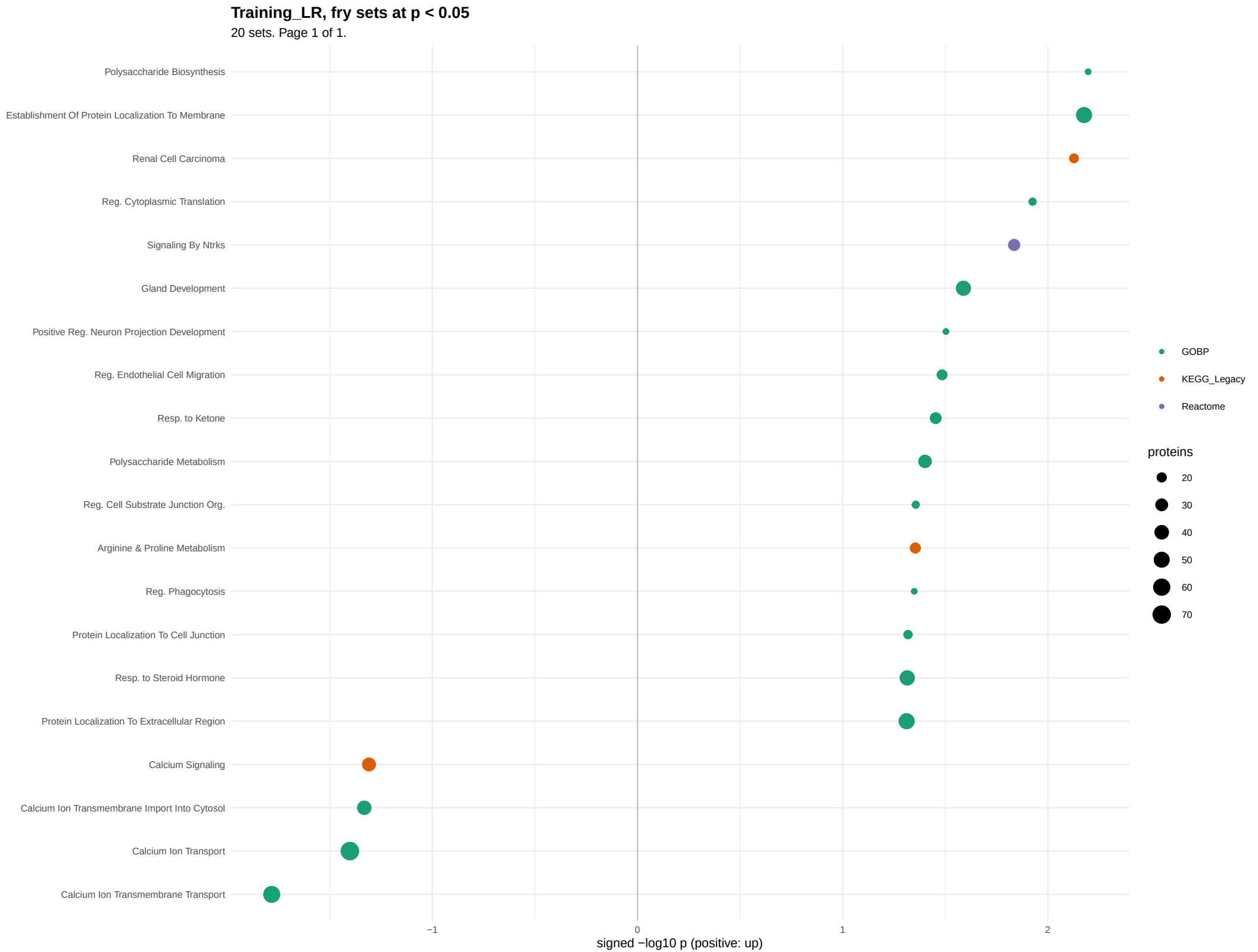

fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

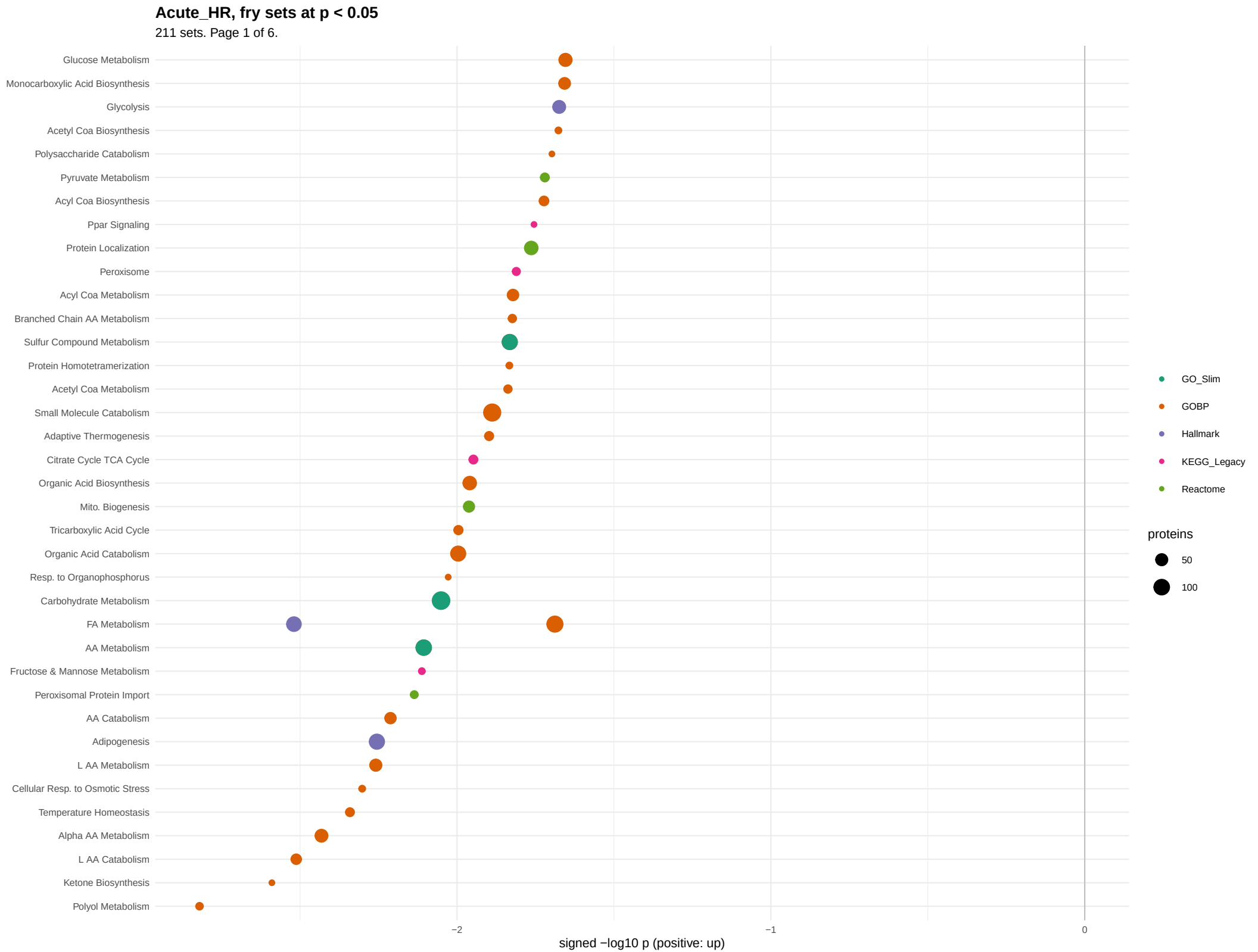

Acute_Interaction, fgsea collapse survivors at FDR < 0.05
63 sets. Page 2 of 2.
Recruitment Of Numa To Mitotic Centrosomes
Cilium Assembly
Resp. to Heat
Reg. Cellular Resp. to Growth Factor Stimulus
UPR
G2M Checkpoint
Multicellular Organismal Movement
Cellular Responses To Mechanical Stimuli
Cell Surface Receptor Protein Serine Threonine Kinase Signaling
Reg. Syncytium Formation By Plasma Membrane Fusion
Mitotic Spindle
Reg. Intracell. Transport
Reg. Cardiac Muscle Contraction
Resp. Syncytial Virus Infection Pathway
MT Polymerization Or Depolymerization
Translational Elongation
Nuclear Transcribed mRNA Catabolism
Negative Reg. Cell Projection Org.
Resp. to Topologically Incorrect Protein
Cytoplasmic Translation
Positive Reg. Protein Localization
Dengue Virus Infection
Eukaryotic Translation Elongation
0.0
0.5
1.0
1.5
2.0
normalised enrichment score
proteins
20
40
60
80
100
GO_Slim
GOBP
Hallmark
Reactome
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

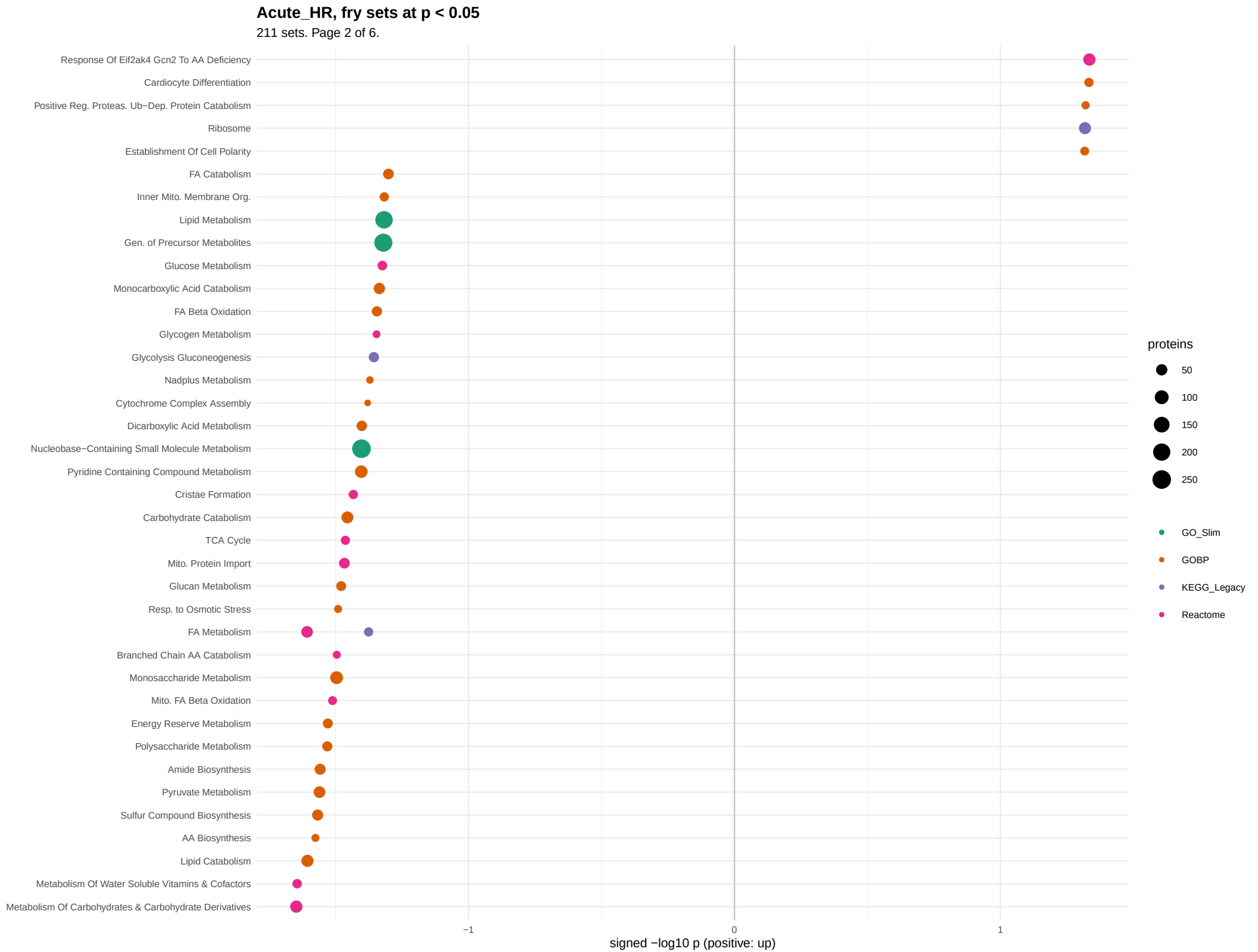


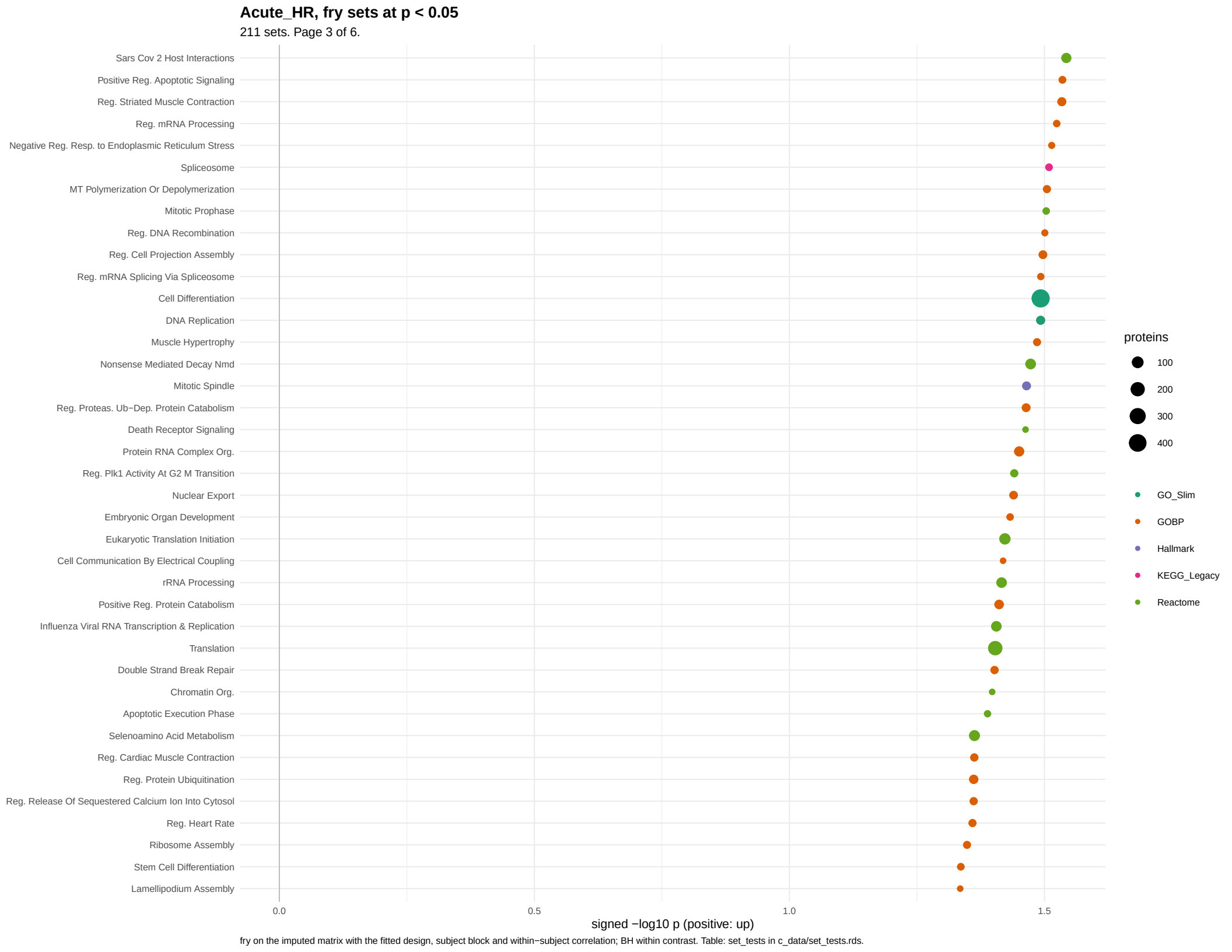

fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

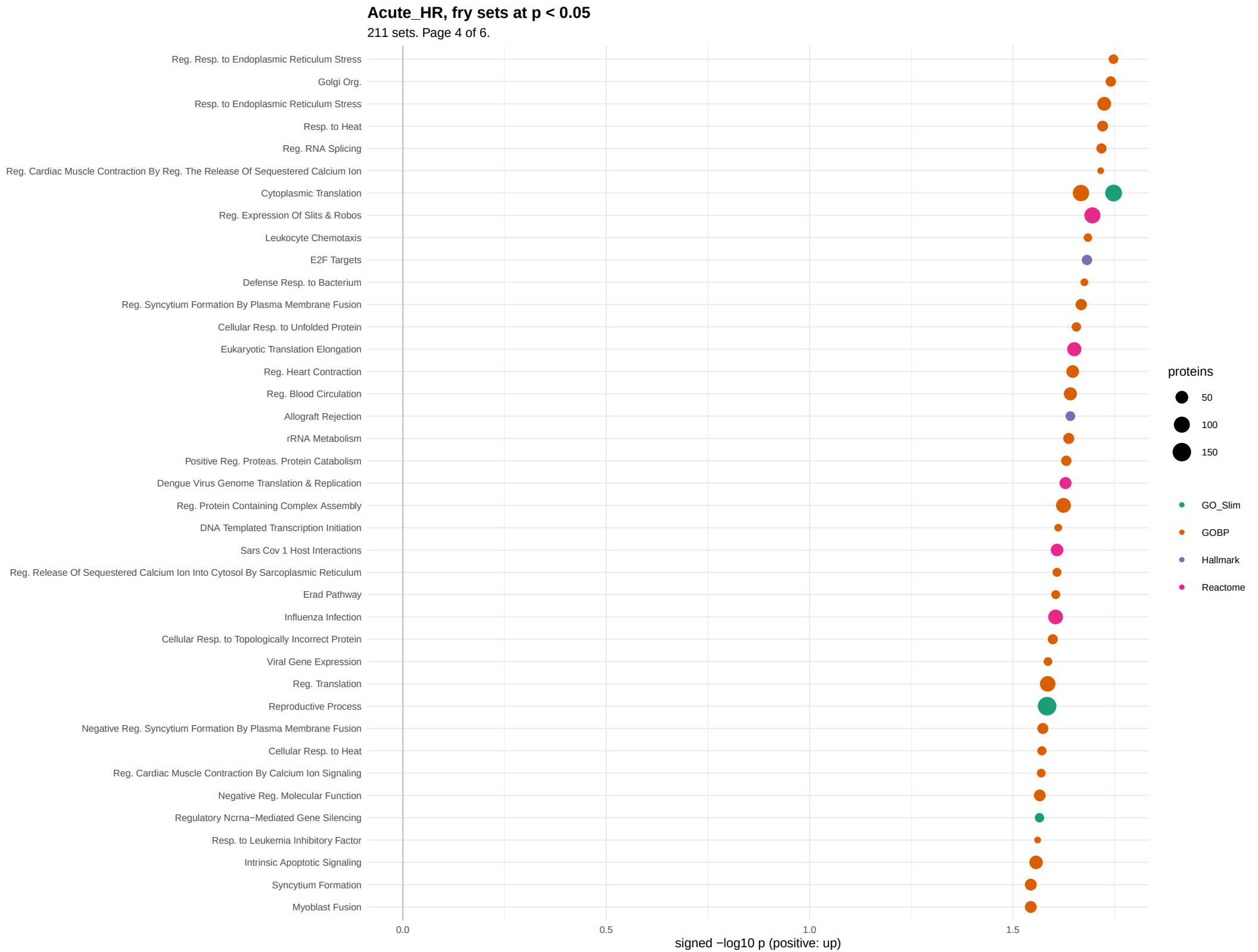

Training_HR, fry sets at p < 0.05
60 sets. Page 2 of 2.
Cellular Resp. to Alcohol
Resp. to Alcohol
Mito. Protein Import
Protein Localization
Learning
AA Activation
Cytochrome Complex Assembly
Ketone Biosynthesis
Protein Homotetramerization
tRNA Aminoacylation
Cell Cycle G1 S Phase Transition
Pallium Development
Positive Reg. Growth
Protein Ubiquitination
Mitotic Prophase
Positive Reg. Actin Filament Bundle Assembly
Heart Morphogenesis
Positive Reg. Cell Development
Resp. to Peptide Hormone
Reg. Muscle Cell Differentiation
−1
0
1
2
signed −log10 p (positive: up)
GOBP
Reactome
proteins
20
40
60
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

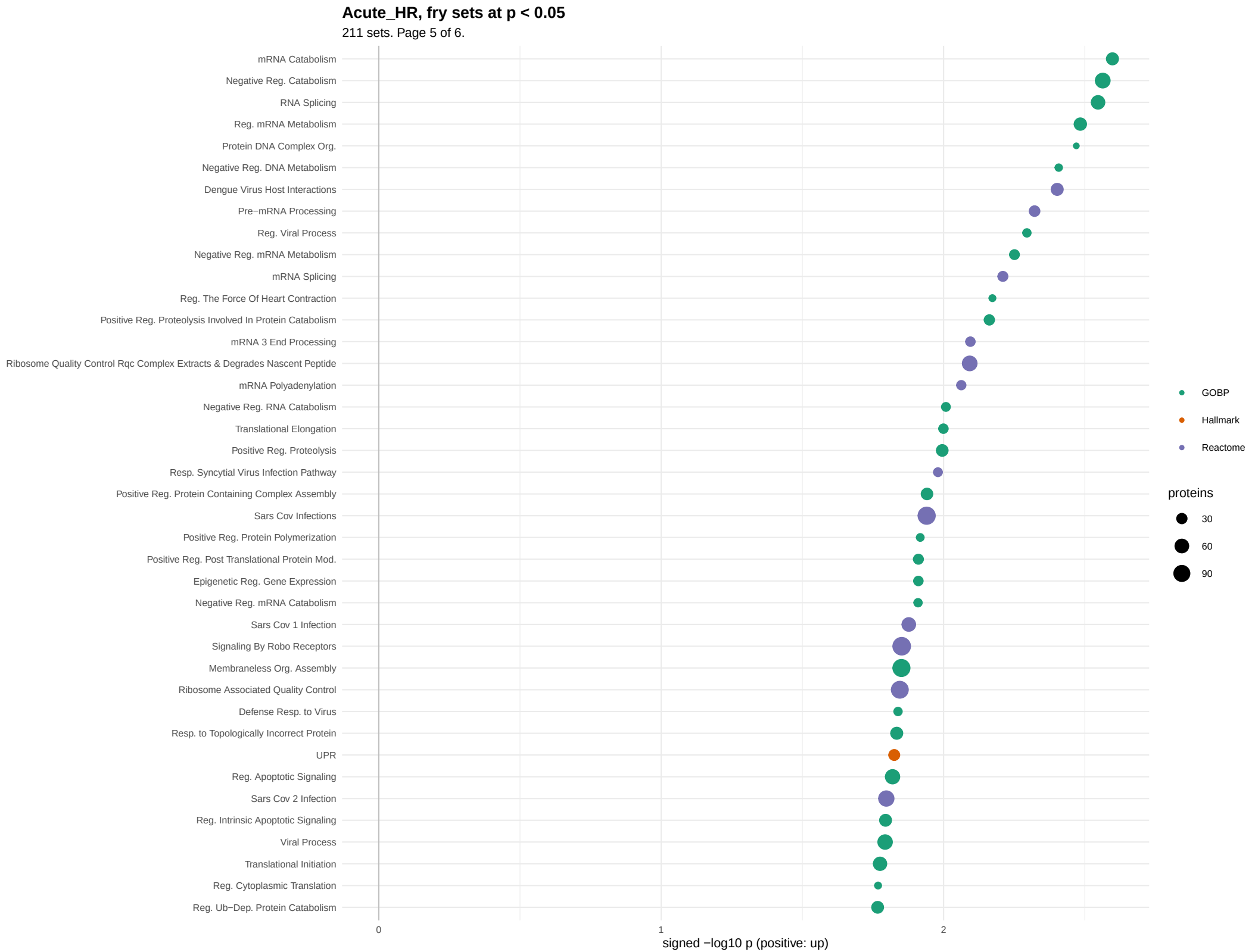

Training_LR, fry sets at p < 0.05
20 sets. Page 1 of 1.
Polysaccharide Biosynthesis
Establishment Of Protein Localization To Membrane
Renal Cell Carcinoma
Reg. Cytoplasmic Translation
Signaling By Ntrks
Gland Development
Positive Reg. Neuron Projection Development
Reg. Endothelial Cell Migration
Resp. to Ketone
Polysaccharide Metabolism
Reg. Cell Substrate Junction Org.
Arginine & Proline Metabolism
Reg. Phagocytosis
Protein Localization To Cell Junction
Resp. to Steroid Hormone
Protein Localization To Extracellular Region
Calcium Signaling
Calcium Ion Transmembrane Import Into Cytosol
Calcium Ion Transport
Calcium Ion Transmembrane Transport
−1
0
1
2
signed −log10 p (positive: up)
GOBP
KEGG_Legacy
Reactome
proteins
20
30
40
50
60
70
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
211 sets. Page 1 of 6.
Glucose Metabolism
Monocarboxylic Acid Biosynthesis
Glycolysis
Acetyl Coa Biosynthesis
Polysaccharide Catabolism
Pyruvate Metabolism
Acyl Coa Biosynthesis
Ppar Signaling
Protein Localization
Peroxisome
Acyl Coa Metabolism
Branched Chain AA Metabolism
Sulfur Compound Metabolism
Protein Homotetramerization
Acetyl Coa Metabolism
Small Molecule Catabolism
Adaptive Thermogenesis
Citrate Cycle TCA Cycle
Organic Acid Biosynthesis
Mito. Biogenesis
Tricarboxylic Acid Cycle
Organic Acid Catabolism
Resp. to Organophosphorus
Carbohydrate Metabolism
FA Metabolism
AA Metabolism
Fructose & Mannose Metabolism
Peroxisomal Protein Import
AA Catabolism
Adipogenesis
L AA Metabolism
Cellular Resp. to Osmotic Stress
Temperature Homeostasis
Alpha AA Metabolism
L AA Catabolism
Ketone Biosynthesis
Polyol Metabolism
−2
−1
0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
211 sets. Page 2 of 6.
Response Of Eif2ak4 Gcn2 To AA Deficiency
Cardiocyte Differentiation
Positive Reg. Proteas. Ub−Dep. Protein Catabolism
Ribosome
Establishment Of Cell Polarity
FA Catabolism
Inner Mito. Membrane Org.
Lipid Metabolism
Gen. of Precursor Metabolites
Glucose Metabolism
Monocarboxylic Acid Catabolism
FA Beta Oxidation
Glycogen Metabolism
Glycolysis Gluconeogenesis
Nadplus Metabolism
Cytochrome Complex Assembly
Dicarboxylic Acid Metabolism
Nucleobase−Containing Small Molecule Metabolism
Pyridine Containing Compound Metabolism
Cristae Formation
Carbohydrate Catabolism
TCA Cycle
Mito. Protein Import
Glucan Metabolism
Resp. to Osmotic Stress
FA Metabolism
Branched Chain AA Catabolism
Monosaccharide Metabolism
Mito. FA Beta Oxidation
Energy Reserve Metabolism
Polysaccharide Metabolism
Amide Biosynthesis
Pyruvate Metabolism
Sulfur Compound Biosynthesis
AA Biosynthesis
Lipid Catabolism
Metabolism Of Water Soluble Vitamins & Cofactors
Metabolism Of Carbohydrates & Carbohydrate Derivatives
−1
0
1
signed −log10 p (positive: up)
proteins
50
100
150
200
250
GO_Slim
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

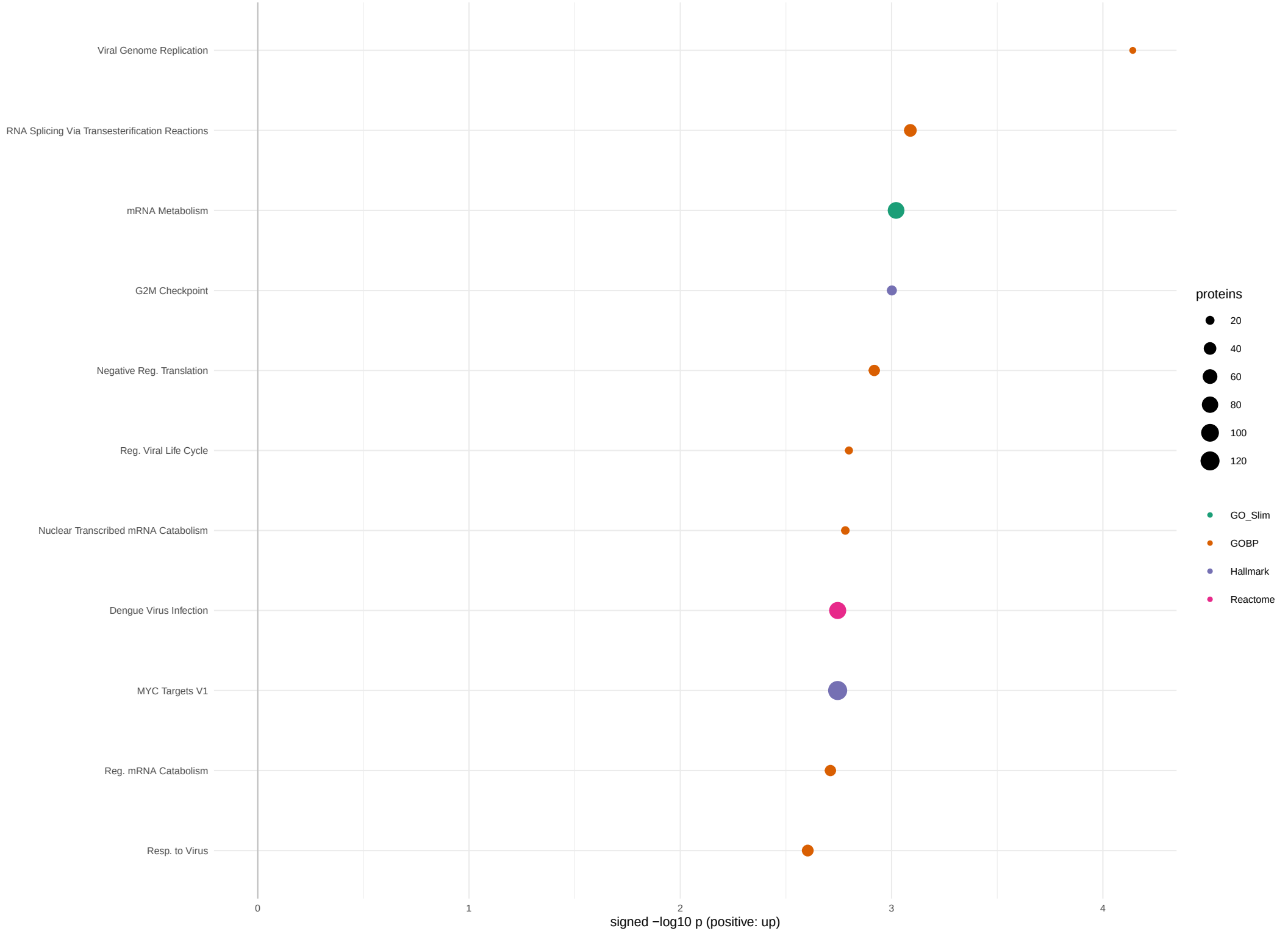

Acute_HR, fry sets at p < 0.05
211 sets. Page 3 of 6.
Sars Cov 2 Host Interactions
Positive Reg. Apoptotic Signaling
Reg. Striated Muscle Contraction
Reg. mRNA Processing
Negative Reg. Resp. to Endoplasmic Reticulum Stress
Spliceosome
MT Polymerization Or Depolymerization
Mitotic Prophase
Reg. DNA Recombination
Reg. Cell Projection Assembly
Reg. mRNA Splicing Via Spliceosome
Cell Differentiation
DNA Replication
Muscle Hypertrophy
Nonsense Mediated Decay Nmd
Mitotic Spindle
Reg. Proteas. Ub−Dep. Protein Catabolism
Death Receptor Signaling
Protein RNA Complex Org.
Reg. Plk1 Activity At G2 M Transition
Nuclear Export
Embryonic Organ Development
Eukaryotic Translation Initiation
Cell Communication By Electrical Coupling
rRNA Processing
Positive Reg. Protein Catabolism
Influenza Viral RNA Transcription & Replication
Translation
Double Strand Break Repair
Chromatin Org.
Apoptotic Execution Phase
Selenoamino Acid Metabolism
Reg. Cardiac Muscle Contraction
Reg. Protein Ubiquitination
Reg. Release Of Sequestered Calcium Ion Into Cytosol
Reg. Heart Rate
Ribosome Assembly
Stem Cell Differentiation
Lamellipodium Assembly
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
100
200
300
400
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
211 sets. Page 4 of 6.
Reg. Resp. to Endoplasmic Reticulum Stress
Golgi Org.
Resp. to Endoplasmic Reticulum Stress
Resp. to Heat
Reg. RNA Splicing
Reg. Cardiac Muscle Contraction By Reg. The Release Of Sequestered Calcium Ion
Cytoplasmic Translation
Reg. Expression Of Slits & Robos
Leukocyte Chemotaxis
E2F Targets
Defense Resp. to Bacterium
Reg. Syncytium Formation By Plasma Membrane Fusion
Cellular Resp. to Unfolded Protein
Eukaryotic Translation Elongation
Reg. Heart Contraction
Reg. Blood Circulation
Allograft Rejection
rRNA Metabolism
Positive Reg. Proteas. Protein Catabolism
Dengue Virus Genome Translation & Replication
Reg. Protein Containing Complex Assembly
DNA Templated Transcription Initiation
Sars Cov 1 Host Interactions
Reg. Release Of Sequestered Calcium Ion Into Cytosol By Sarcoplasmic Reticulum
Erad Pathway
Influenza Infection
Cellular Resp. to Topologically Incorrect Protein
Viral Gene Expression
Reg. Translation
Reproductive Process
Negative Reg. Syncytium Formation By Plasma Membrane Fusion
Cellular Resp. to Heat
Reg. Cardiac Muscle Contraction By Calcium Ion Signaling
Negative Reg. Molecular Function
Regulatory Ncrna−Mediated Gene Silencing
Resp. to Leukemia Inhibitory Factor
Intrinsic Apoptotic Signaling
Syncytium Formation
Myoblast Fusion
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
50
100
150
GO_Slim
GOBP
Hallmark
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

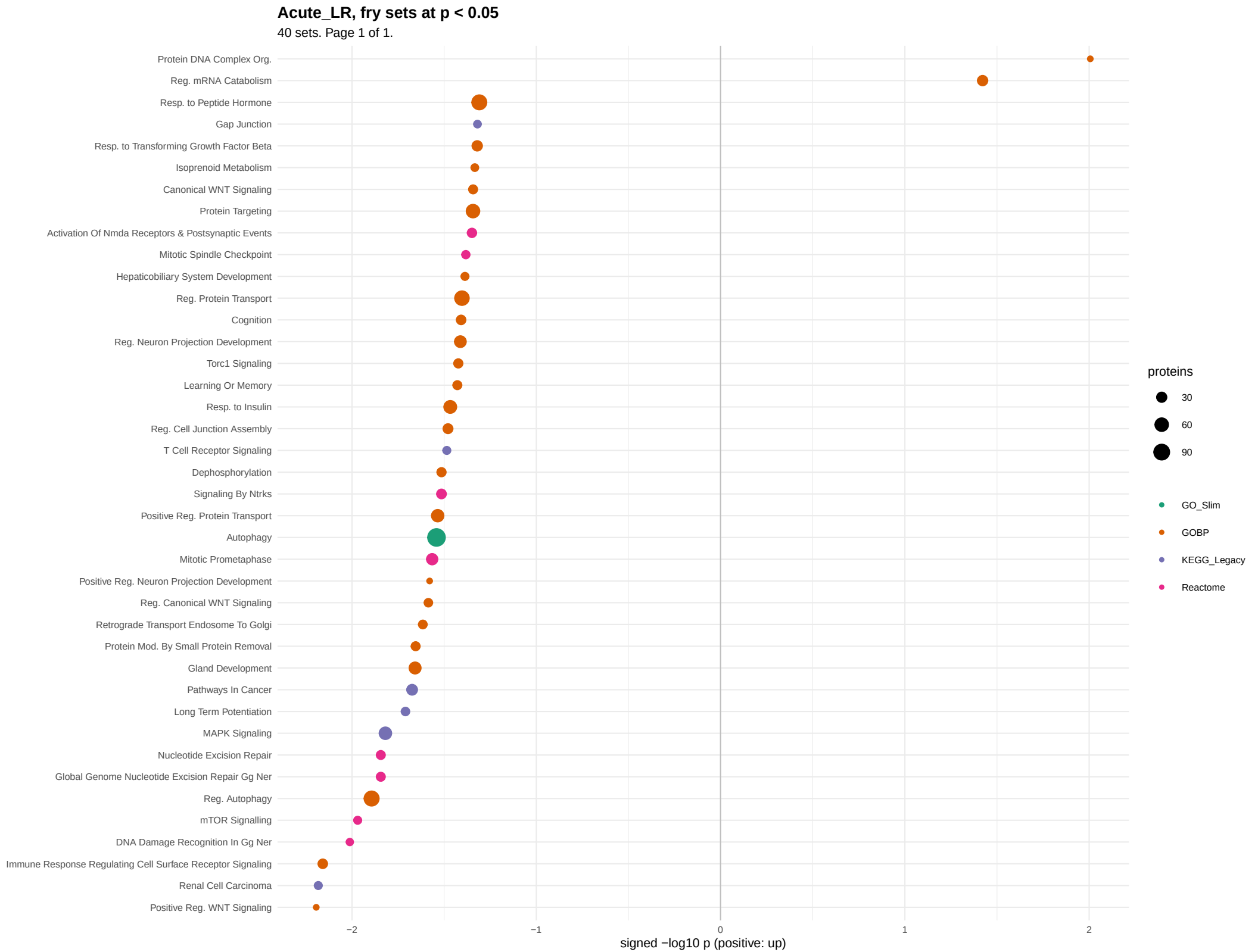

Acute_HR, fry sets at p < 0.05
211 sets. Page 5 of 6.
mRNA Catabolism
Negative Reg. Catabolism
RNA Splicing
Reg. mRNA Metabolism
Protein DNA Complex Org.
Negative Reg. DNA Metabolism
Dengue Virus Host Interactions
Pre−mRNA Processing
Reg. Viral Process
Negative Reg. mRNA Metabolism
mRNA Splicing
Reg. The Force Of Heart Contraction
Positive Reg. Proteolysis Involved In Protein Catabolism
mRNA 3 End Processing
Ribosome Quality Control Rqc Complex Extracts & Degrades Nascent Peptide
mRNA Polyadenylation
Negative Reg. RNA Catabolism
Translational Elongation
Positive Reg. Proteolysis
Resp. Syncytial Virus Infection Pathway
Positive Reg. Protein Containing Complex Assembly
Sars Cov Infections
Positive Reg. Protein Polymerization
Positive Reg. Post Translational Protein Mod.
Epigenetic Reg. Gene Expression
Negative Reg. mRNA Catabolism
Sars Cov 1 Infection
Signaling By Robo Receptors
Membraneless Org. Assembly
Ribosome Associated Quality Control
Defense Resp. to Virus
Resp. to Topologically Incorrect Protein
UPR
Reg. Apoptotic Signaling
Sars Cov 2 Infection
Reg. Intrinsic Apoptotic Signaling
Viral Process
Translational Initiation
Reg. Cytoplasmic Translation
Reg. Ub−Dep. Protein Catabolism
0
1
2
signed −log10 p (positive: up)
GOBP
Hallmark
Reactome
proteins
30
60
90
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

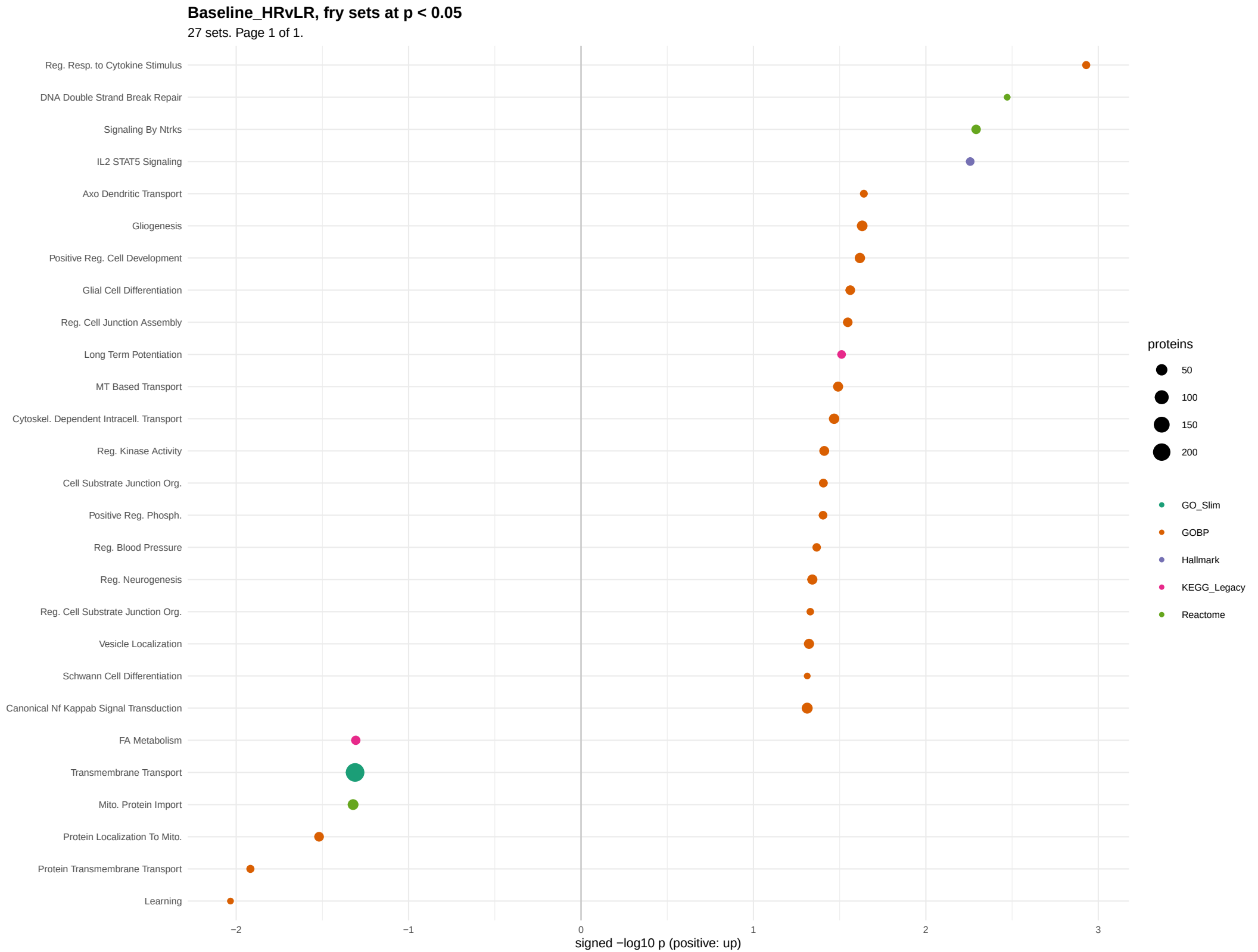
**Acute_HR, fry sets at p < 0.05**

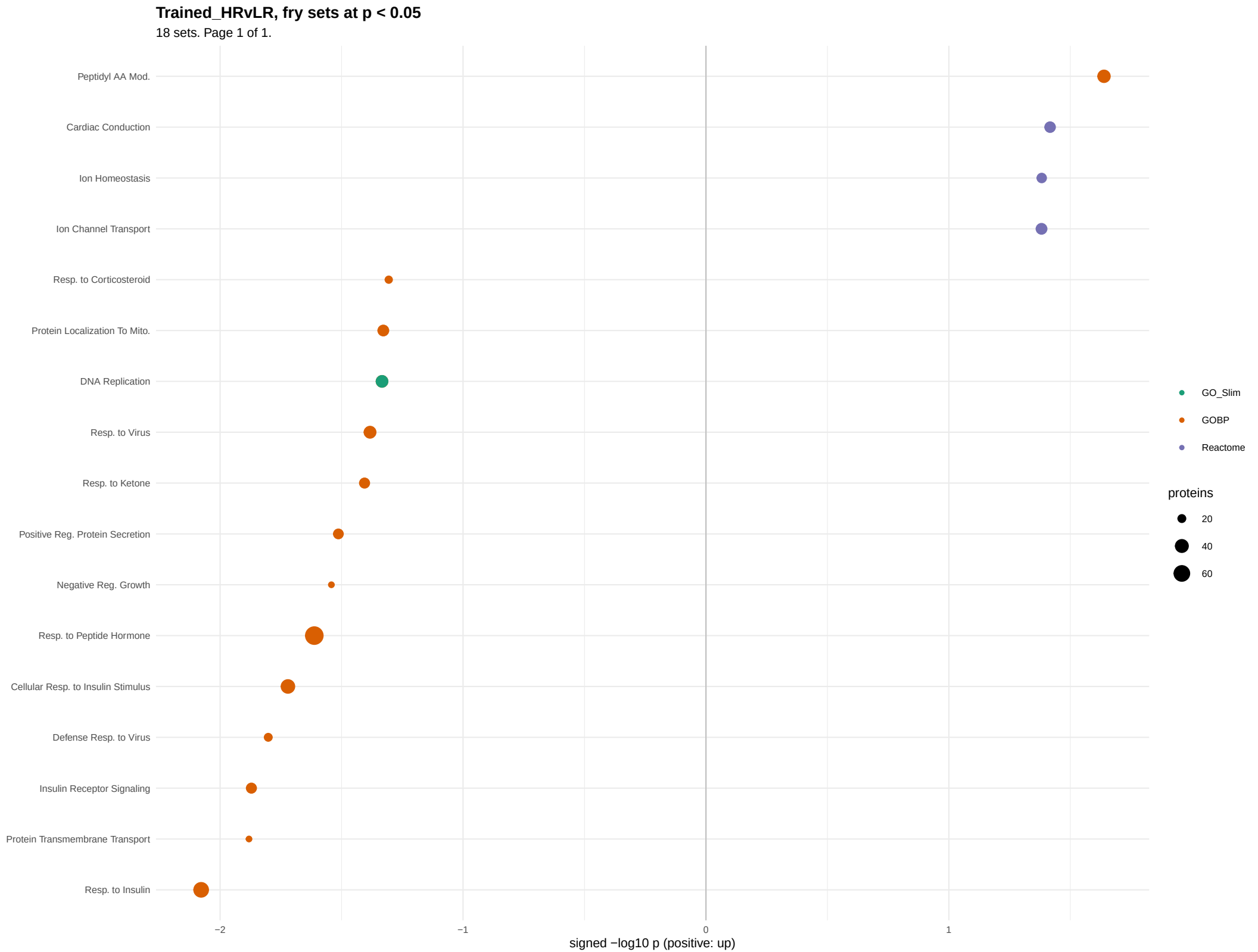
211 sets. Page 6 of 6.

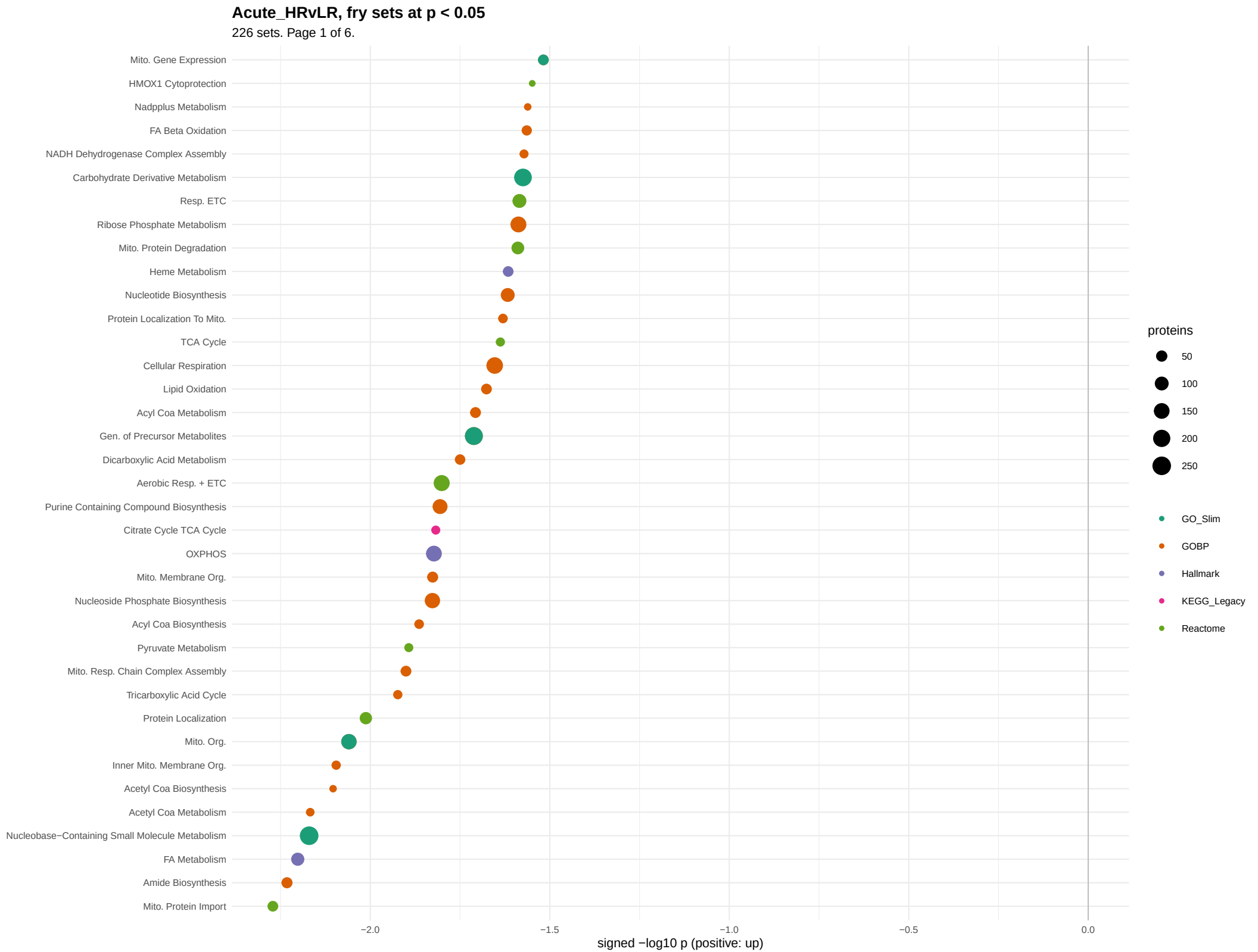



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

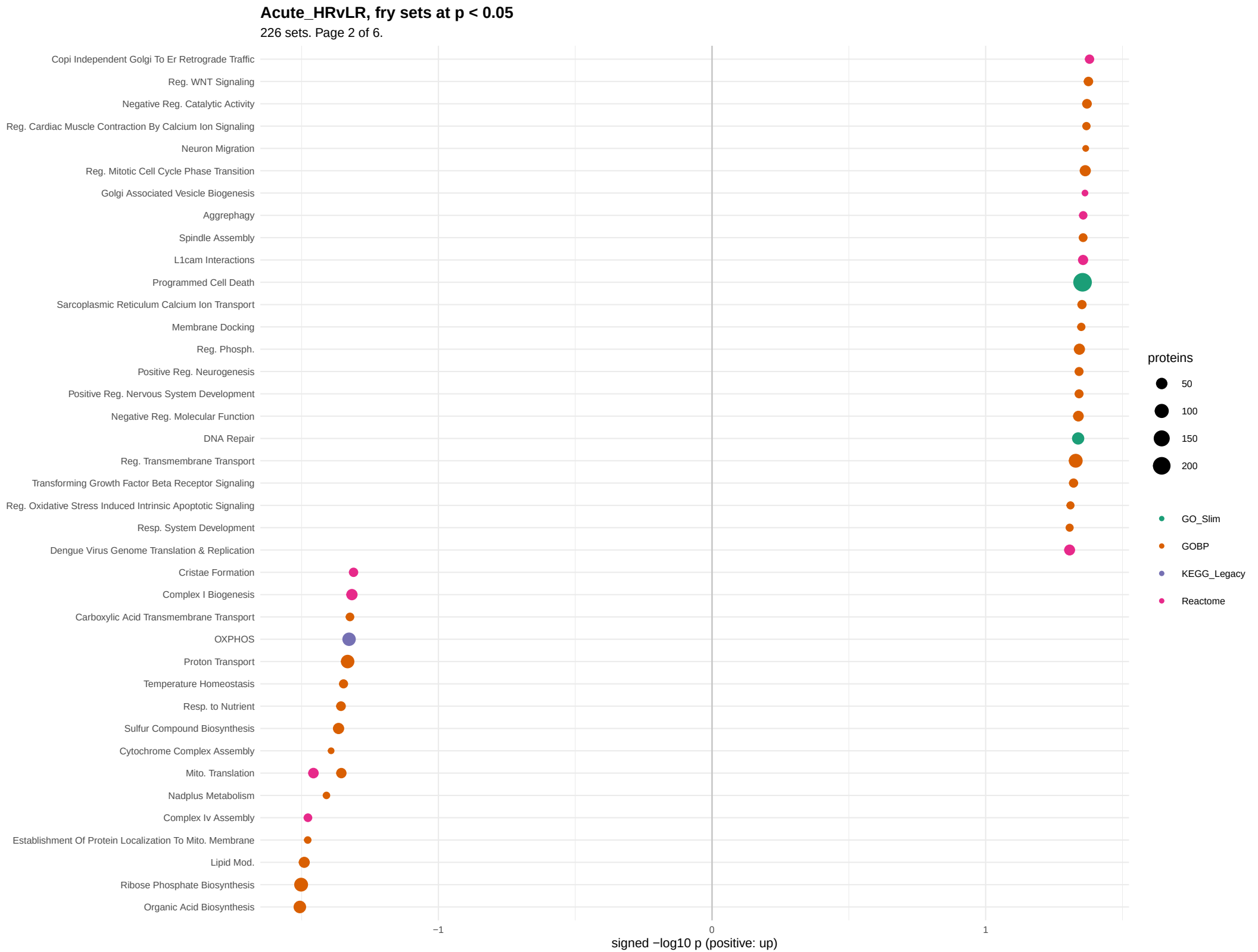

Acute_LR, fry sets at p < 0.05
40 sets. Page 1 of 1.
Protein DNA Complex Org.
Reg. mRNA Catabolism
Resp. to Peptide Hormone
Gap Junction
Resp. to Transforming Growth Factor Beta
Isoprenoid Metabolism
Canonical WNT Signaling
Protein Targeting
Activation Of Nmda Receptors & Postsynaptic Events
Mitotic Spindle Checkpoint
Hepaticobiliary System Development
Reg. Protein Transport
Cognition
Reg. Neuron Projection Development
Torc1 Signaling
Learning Or Memory
Resp. to Insulin
Reg. Cell Junction Assembly
T Cell Receptor Signaling
Dephosphorylation
Signaling By Ntrks
Positive Reg. Protein Transport
Autophagy
Mitotic Prometaphase
Positive Reg. Neuron Projection Development
Reg. Canonical WNT Signaling
Retrograde Transport Endosome To Golgi
Protein Mod. By Small Protein Removal
Gland Development
Pathways In Cancer
Long Term Potentiation
MAPK Signaling
Nucleotide Excision Repair
Global Genome Nucleotide Excision Repair Gg Ner
Reg. Autophagy
mTOR Signalling
DNA Damage Recognition In Gg Ner
Immune Response Regulating Cell Surface Receptor Signaling
Renal Cell Carcinoma
Positive Reg. WNT Signaling
−2
−1
0
1
2
signed −log10 p (positive: up)
proteins
30
60
90
GO_Slim
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

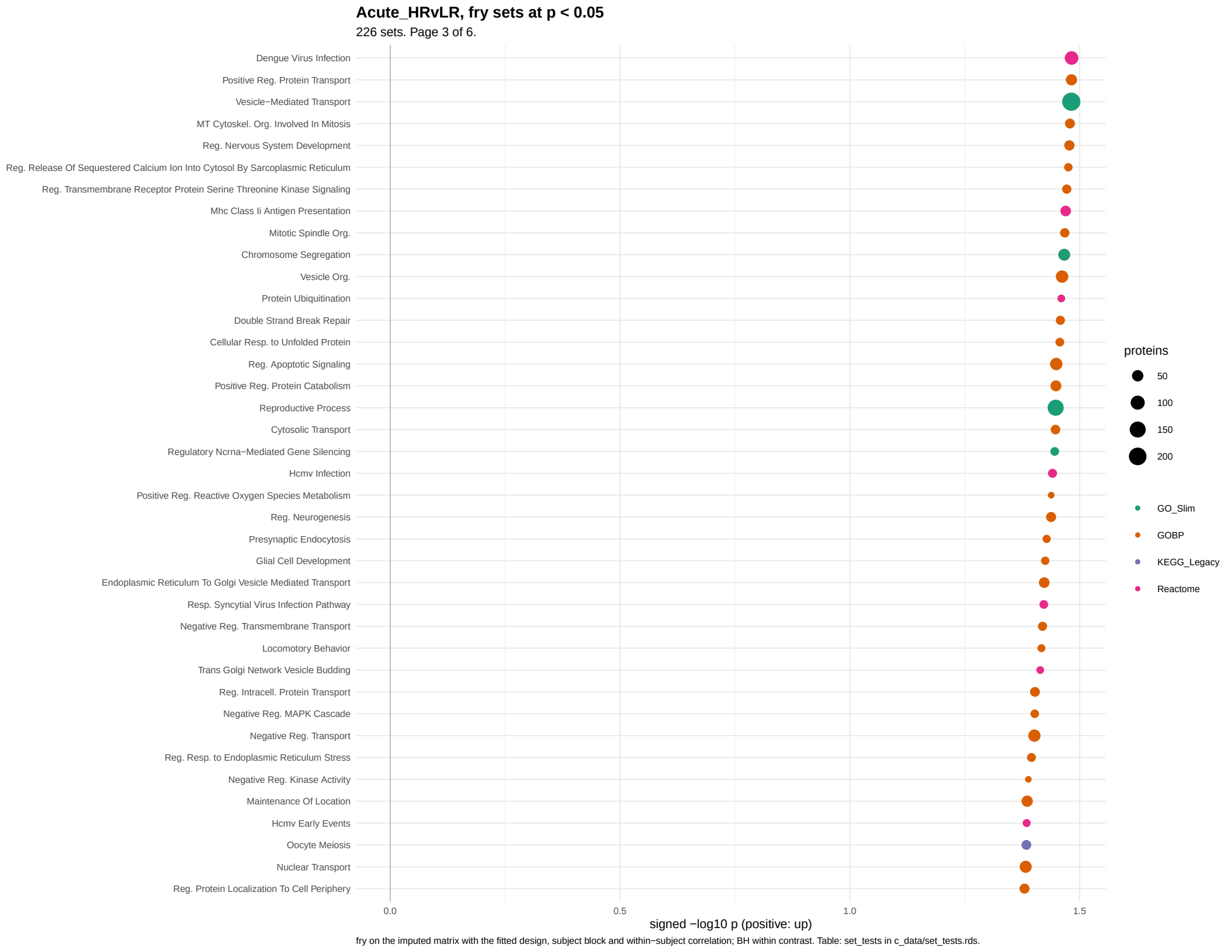

Baseline_HRvLR, fry sets at p < 0.05
27 sets. Page 1 of 1.
Reg. Resp. to Cytokine Stimulus
DNA Double Strand Break Repair
Signaling By Ntrks
IL2 STAT5 Signaling
Axo Dendritic Transport
Gliogenesis
Positive Reg. Cell Development
Glial Cell Differentiation
Reg. Cell Junction Assembly
Long Term Potentiation
MT Based Transport
Cytoskel. Dependent Intracell. Transport
Reg. Kinase Activity
Cell Substrate Junction Org.
Positive Reg. Phosph.
Reg. Blood Pressure
Reg. Neurogenesis
Reg. Cell Substrate Junction Org.
Vesicle Localization
Schwann Cell Differentiation
Canonical Nf Kappab Signal Transduction
FA Metabolism
Transmembrane Transport
Mito. Protein Import
Protein Localization To Mito.
Protein Transmembrane Transport
Learning
−2
−1
0
1
2
3
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Trained_HRvLR, fry sets at p < 0.05
18 sets. Page 1 of 1.
Peptidyl AA Mod.
Cardiac Conduction
Ion Homeostasis
Ion Channel Transport
Resp. to Corticosteroid
Protein Localization To Mito.
DNA Replication
Resp. to Virus
Resp. to Ketone
Positive Reg. Protein Secretion
Negative Reg. Growth
Resp. to Peptide Hormone
Cellular Resp. to Insulin Stimulus
Defense Resp. to Virus
Insulin Receptor Signaling
Protein Transmembrane Transport
Resp. to Insulin
−2
−1
0
1
signed −log10 p (positive: up)
GO_Slim
GOBP
Reactome
proteins
20
40
60
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



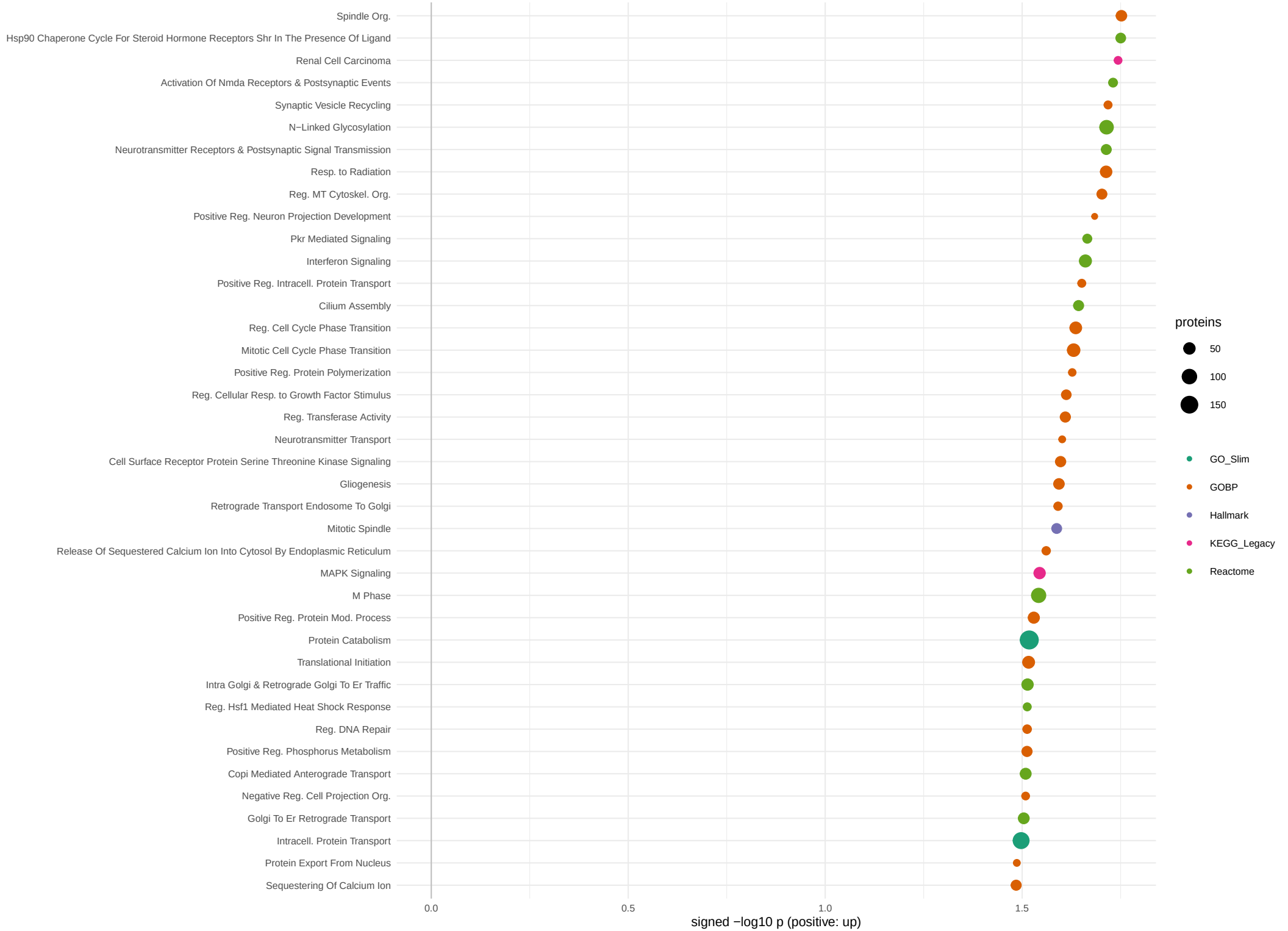

fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HRvLR, fry sets at p < 0.05
226 sets. Page 2 of 6.
Copi Independent Golgi To Er Retrograde Traffic
Reg. WNT Signaling
Negative Reg. Catalytic Activity
Reg. Cardiac Muscle Contraction By Calcium Ion Signaling
Neuron Migration
Reg. Mitotic Cell Cycle Phase Transition
Golgi Associated Vesicle Biogenesis
Aggrephagy
Spindle Assembly
L1cam Interactions
Programmed Cell Death
Sarcoplasmic Reticulum Calcium Ion Transport
Membrane Docking
Reg. Phosph.
Positive Reg. Neurogenesis
Positive Reg. Nervous System Development
Negative Reg. Molecular Function
DNA Repair
Reg. Transmembrane Transport
Transforming Growth Factor Beta Receptor Signaling
Reg. Oxidative Stress Induced Intrinsic Apoptotic Signaling
Resp. System Development
Dengue Virus Genome Translation & Replication
Cristae Formation
Complex I Biogenesis
Carboxylic Acid Transmembrane Transport
OXPHOS
Proton Transport
Temperature Homeostasis
Resp. to Nutrient
Sulfur Compound Biosynthesis
Cytochrome Complex Assembly
Mito. Translation
Nadplus Metabolism
Complex Iv Assembly
Establishment Of Protein Localization To Mito. Membrane
Lipid Mod.
Ribose Phosphate Biosynthesis
Organic Acid Biosynthesis
−1
0
1
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HRvLR, fry sets at p < 0.05
226 sets. Page 3 of 6.
Dengue Virus Infection
Positive Reg. Protein Transport
Vesicle−Mediated Transport
MT Cytoskel. Org. Involved In Mitosis
Reg. Nervous System Development
Reg. Release Of Sequestered Calcium Ion Into Cytosol By Sarcoplasmic Reticulum
Reg. Transmembrane Receptor Protein Serine Threonine Kinase Signaling
Mhc Class Ii Antigen Presentation
Mitotic Spindle Org.
Chromosome Segregation
Vesicle Org.
Protein Ubiquitination
Double Strand Break Repair
Cellular Resp. to Unfolded Protein
Reg. Apoptotic Signaling
Positive Reg. Protein Catabolism
Reproductive Process
Cytosolic Transport
Regulatory Ncrna−Mediated Gene Silencing
Hcmv Infection
Positive Reg. Reactive Oxygen Species Metabolism
Reg. Neurogenesis
Presynaptic Endocytosis
Glial Cell Development
Endoplasmic Reticulum To Golgi Vesicle Mediated Transport
Resp. Syncytial Virus Infection Pathway
Negative Reg. Transmembrane Transport
Locomotory Behavior
Trans Golgi Network Vesicle Budding
Reg. Intracell. Protein Transport
Negative Reg. MAPK Cascade
Negative Reg. Transport
Reg. Resp. to Endoplasmic Reticulum Stress
Negative Reg. Kinase Activity
Maintenance Of Location
Hcmv Early Events
Oocyte Meiosis
Nuclear Transport
Reg. Protein Localization To Cell Periphery
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

## Acute_HRvLR, fry sets at p < 0.05

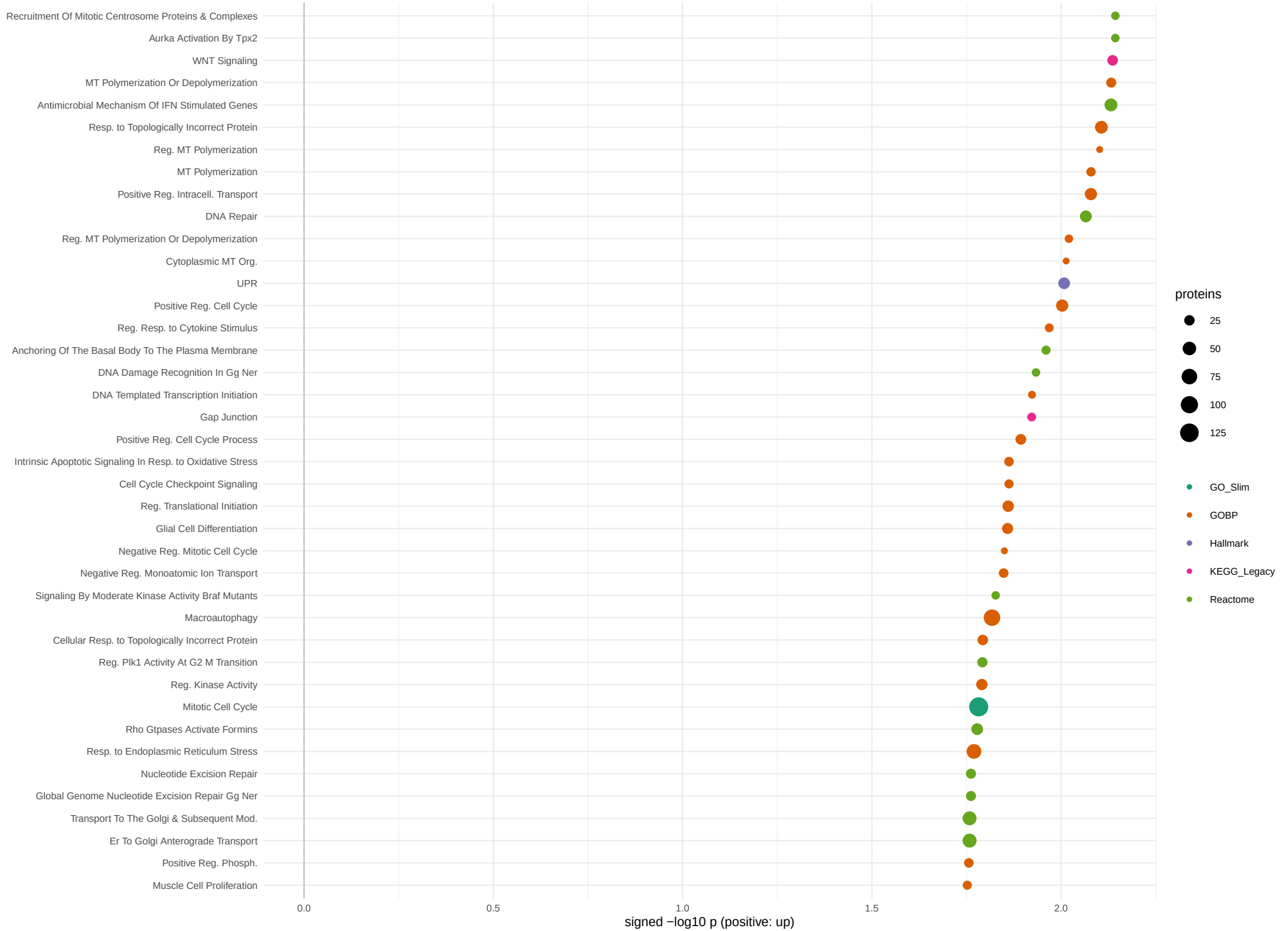
226 sets. Page 4 of 6.



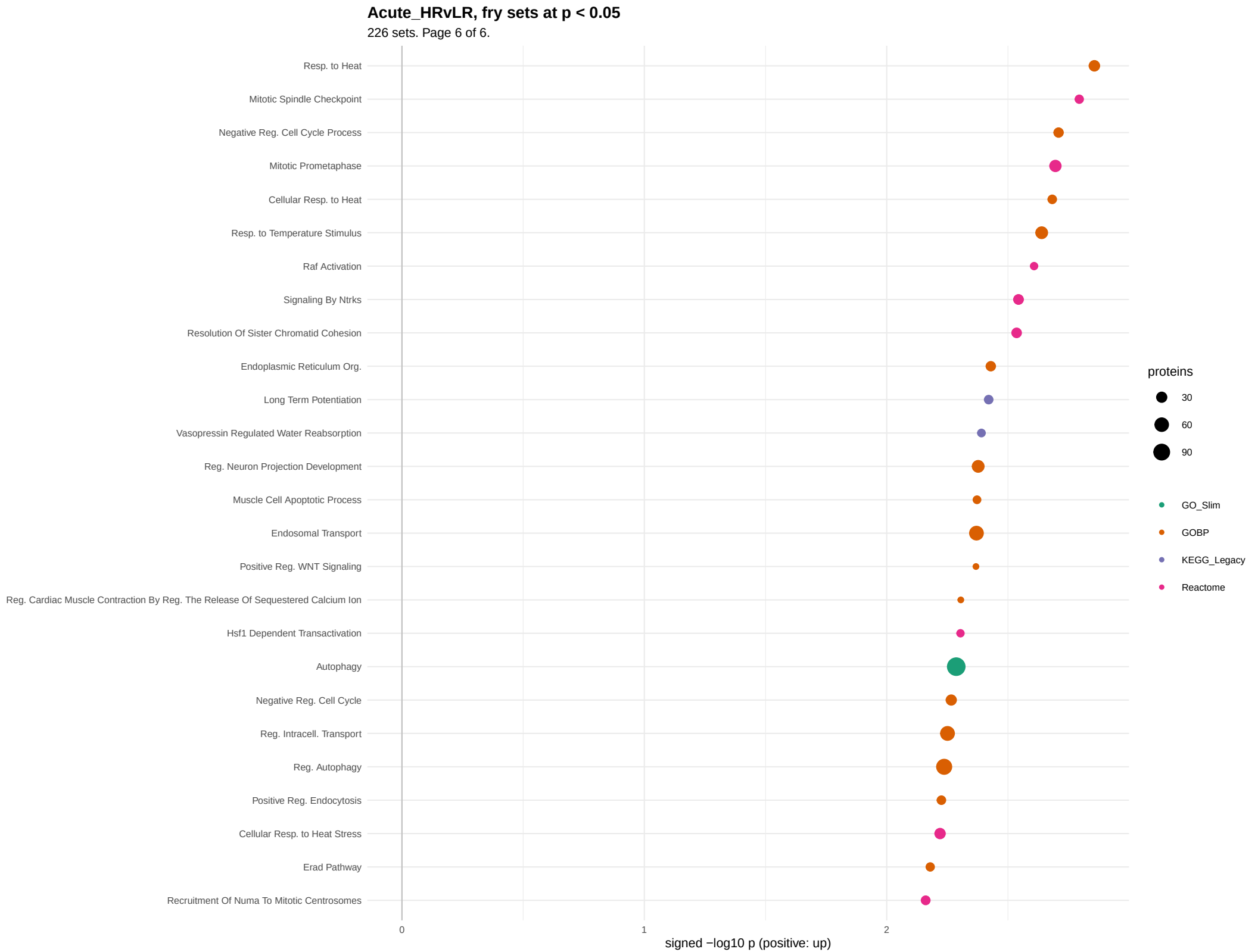

fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

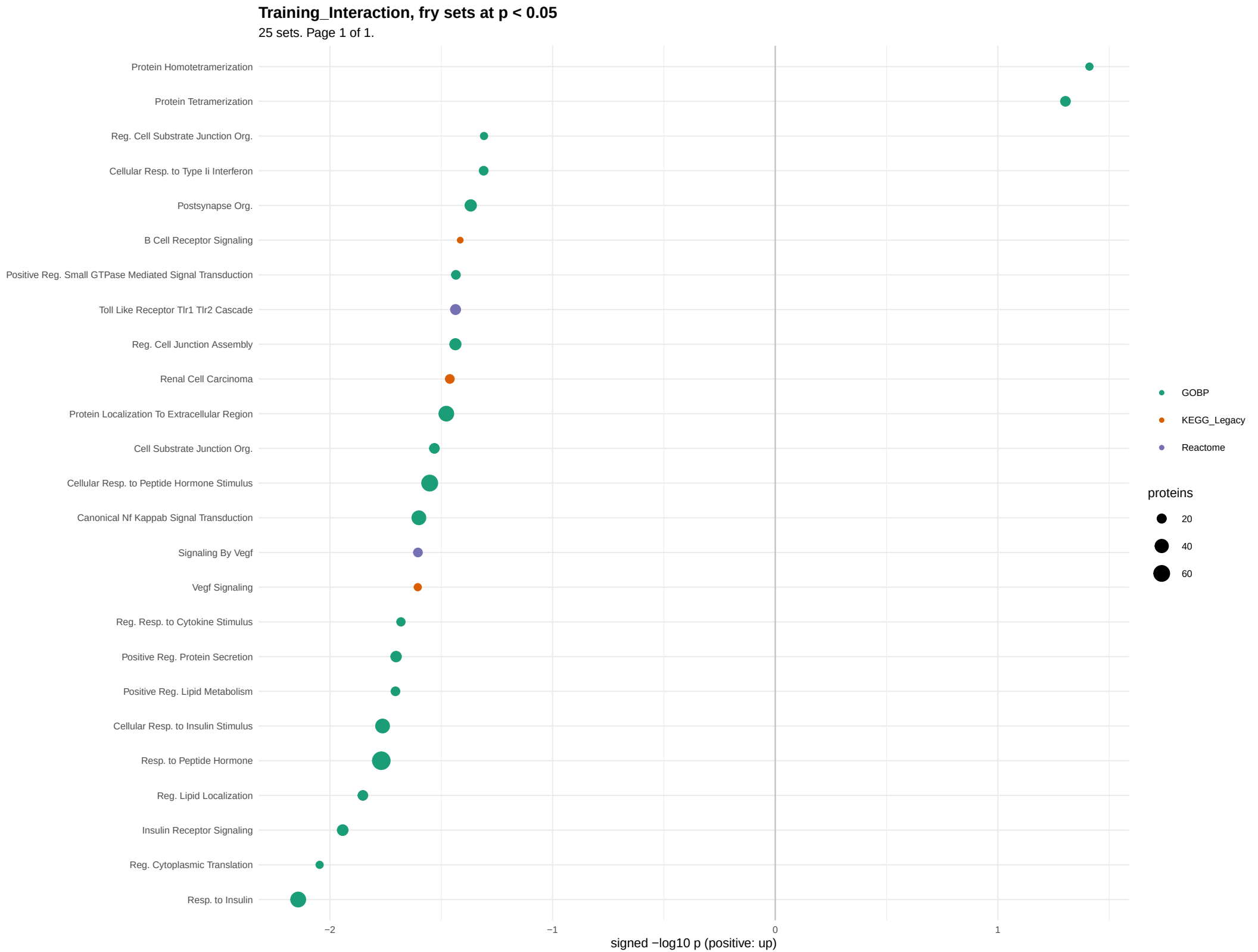
# Acute_HRvLR, fry sets at p < 0.05

226 sets. Page 5 of 6.

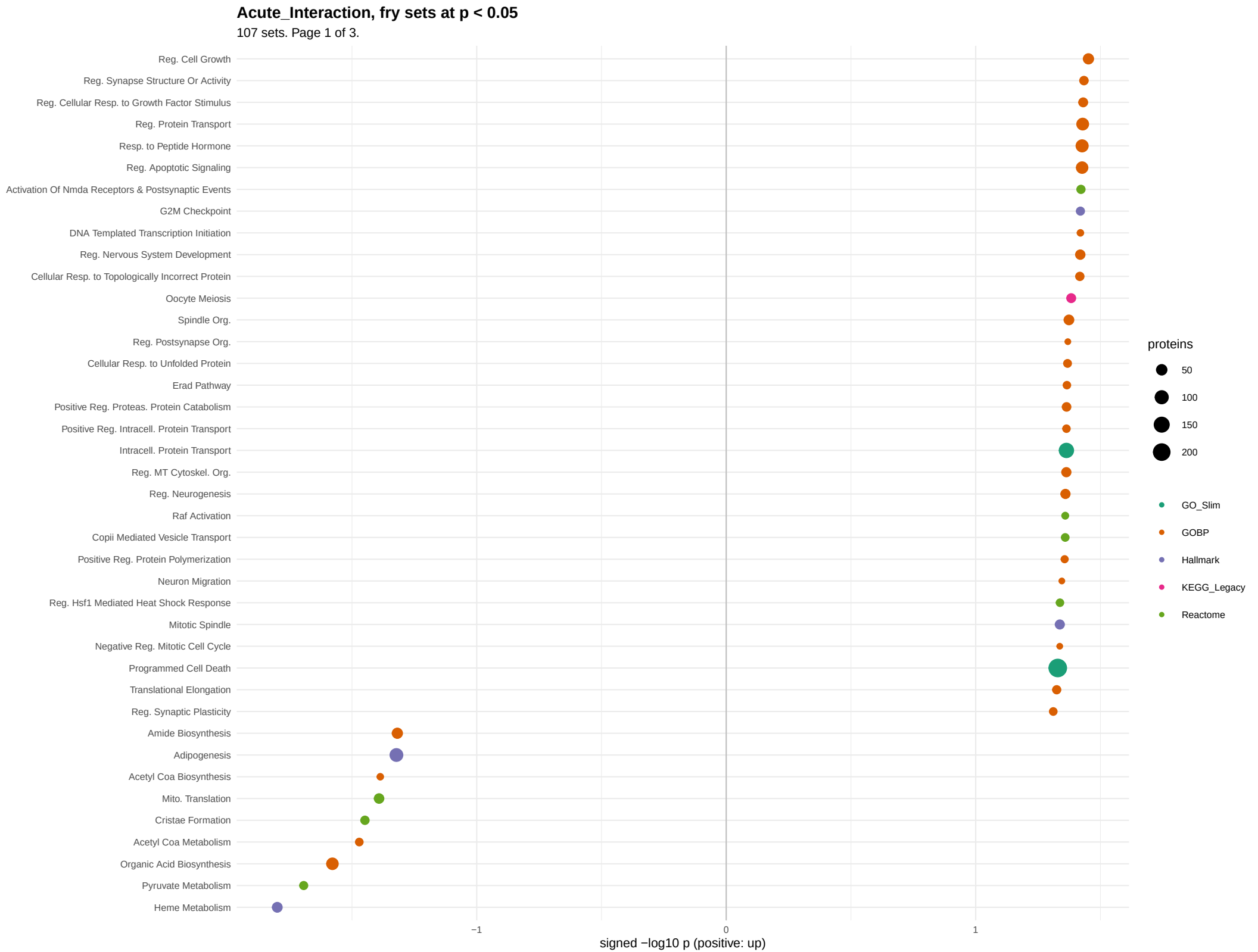


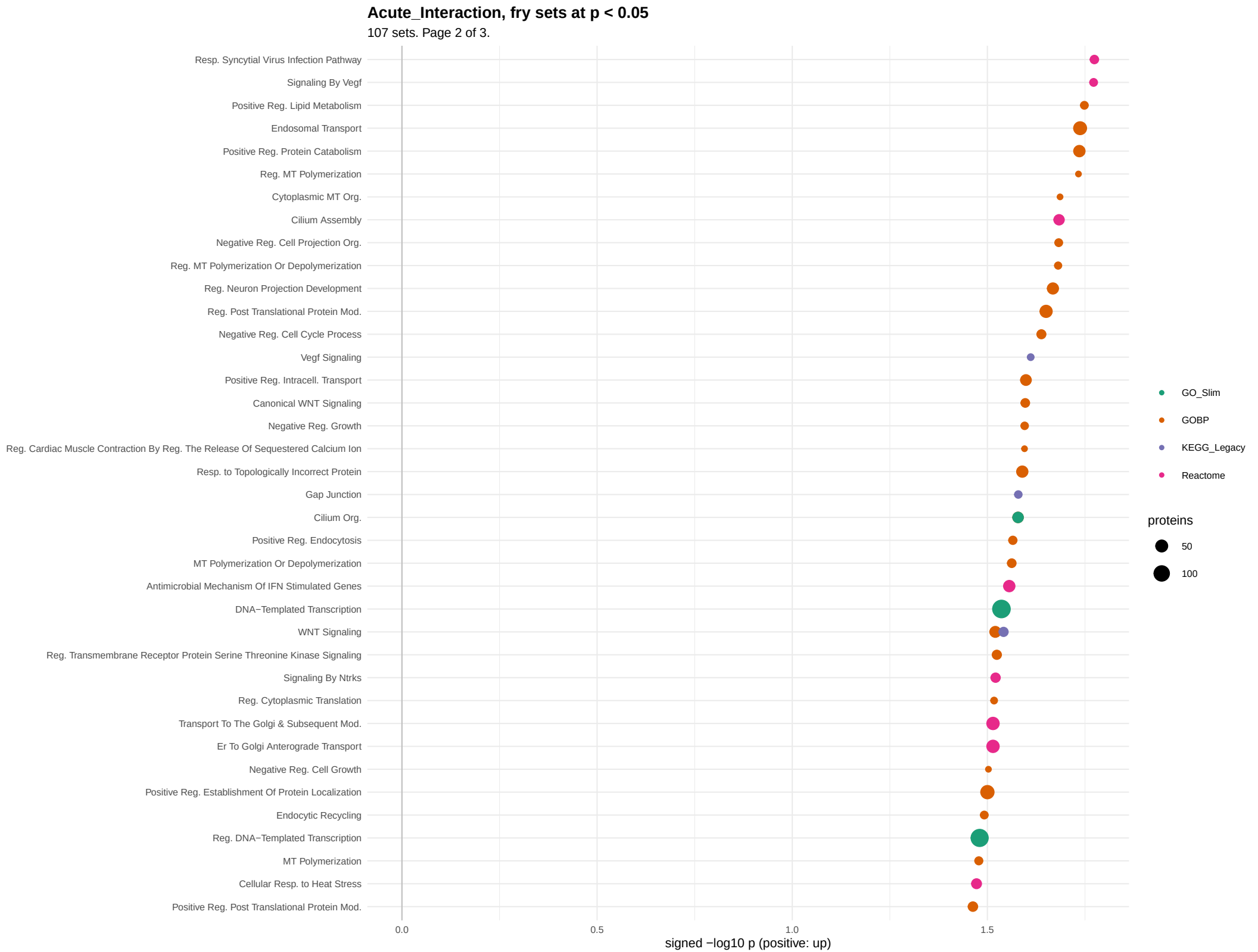

fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

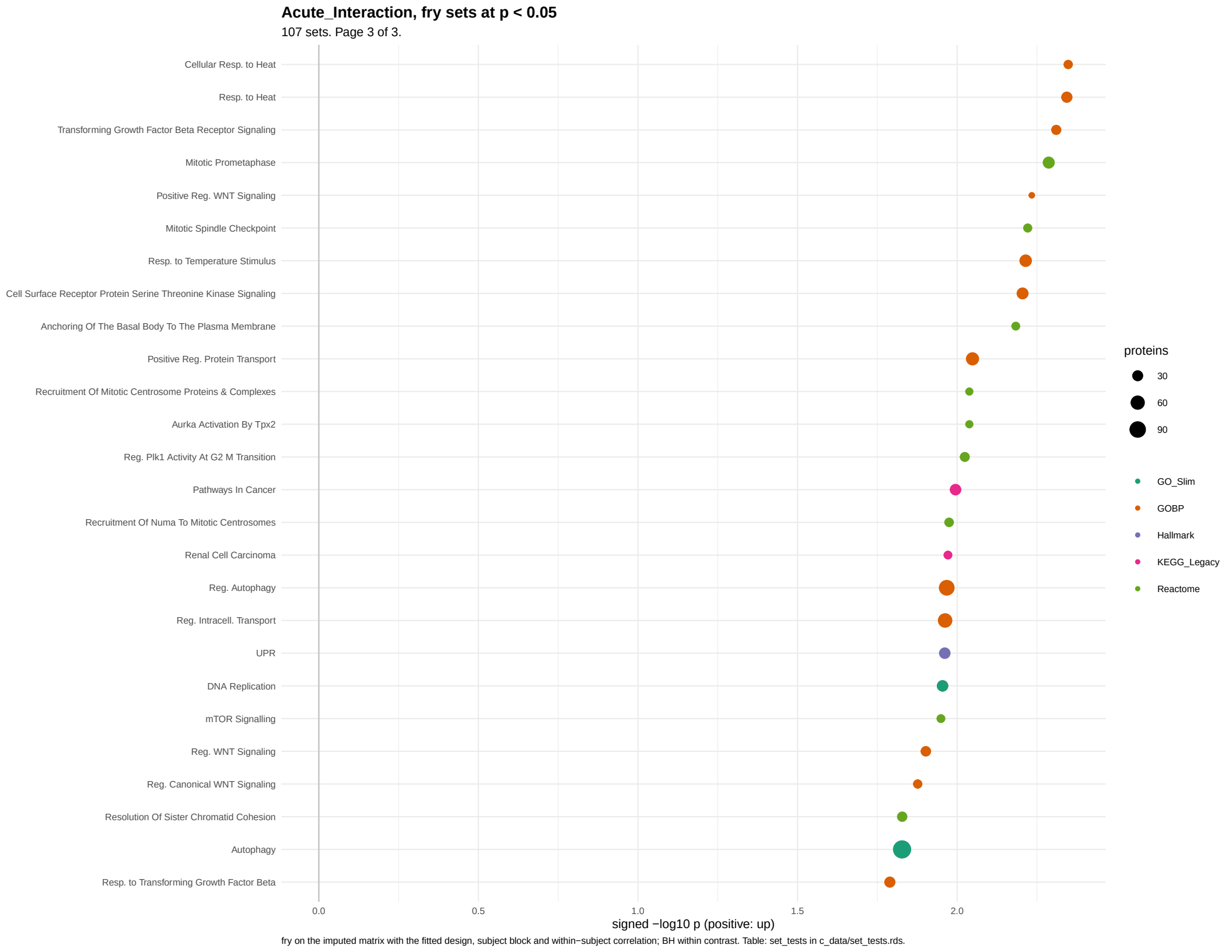

Acute_HRvLR, fry sets at p < 0.05
226 sets. Page 6 of 6.
Resp. to Heat
Mitotic Spindle Checkpoint
Negative Reg. Cell Cycle Process
Mitotic Prometaphase
Cellular Resp. to Heat
Resp. to Temperature Stimulus
Raf Activation
Signaling By Ntrks
Resolution Of Sister Chromatid Cohesion
Endoplasmic Reticulum Org.
Long Term Potentiation
Vasopressin Regulated Water Reabsorption
Reg. Neuron Projection Development
Muscle Cell Apoptotic Process
Endosomal Transport
Positive Reg. WNT Signaling
Reg. Cardiac Muscle Contraction By Reg. The Release Of Sequestered Calcium Ion
Hsf1 Dependent Transactivation
Autophagy
Negative Reg. Cell Cycle
Reg. Intracell. Transport
Reg. Autophagy
Positive Reg. Endocytosis
Cellular Resp. to Heat Stress
Erad Pathway
Recruitment Of Numa To Mitotic Centrosomes
0
1
2
signed −log10 p (positive: up)
proteins
30
60
90
GO_Slim
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

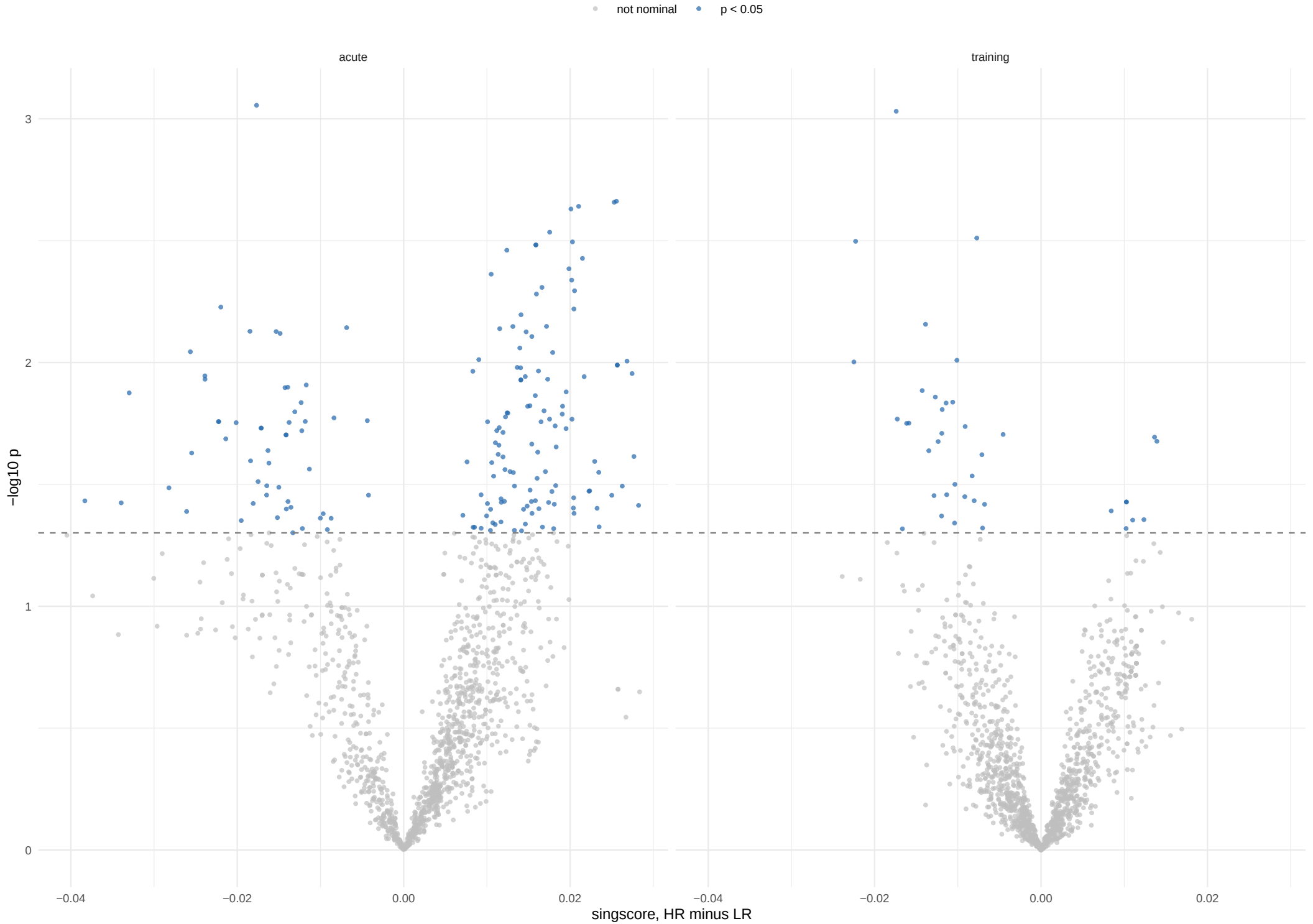

Acute_Interaction, fry sets at p < 0.05
107 sets. Page 1 of 3.
Reg. Cell Growth
Reg. Synapse Structure Or Activity
Reg. Cellular Resp. to Growth Factor Stimulus
Reg. Protein Transport
Resp. to Peptide Hormone
Reg. Apoptotic Signaling
Activation Of Nmda Receptors & Postsynaptic Events
G2M Checkpoint
DNA Templated Transcription Initiation
Reg. Nervous System Development
Cellular Resp. to Topologically Incorrect Protein
Oocyte Meiosis
Spindle Org.
Reg. Postsynapse Org.
Cellular Resp. to Unfolded Protein
Erad Pathway
Positive Reg. Proteas. Protein Catabolism
Positive Reg. Intracell. Protein Transport
Intracell. Protein Transport
Reg. MT Cytoskel. Org.
Reg. Neurogenesis
Raf Activation
Copii Mediated Vesicle Transport
Positive Reg. Protein Polymerization
Neuron Migration
Reg. Hsf1 Mediated Heat Shock Response
Mitotic Spindle
Negative Reg. Mitotic Cell Cycle
Programmed Cell Death
Translational Elongation
Reg. Synaptic Plasticity
Amide Biosynthesis
Adipogenesis
Acetyl Coa Biosynthesis
Mito. Translation
Cristae Formation
Acetyl Coa Metabolism
Organic Acid Biosynthesis
Pyruvate Metabolism
Heme Metabolism
−1
0
1
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_Interaction, fry sets at p < 0.05
107 sets. Page 2 of 3.
Resp. Syncytial Virus Infection Pathway
Signaling By Vegf
Positive Reg. Lipid Metabolism
Endosomal Transport
Positive Reg. Protein Catabolism
Reg. MT Polymerization
Cytoplasmic MT Org.
Cilium Assembly
Negative Reg. Cell Projection Org.
Reg. MT Polymerization Or Depolymerization
Reg. Neuron Projection Development
Reg. Post Translational Protein Mod.
Negative Reg. Cell Cycle Process
Vegf Signaling
Positive Reg. Intracell. Transport
Canonical WNT Signaling
Negative Reg. Growth
Reg. Cardiac Muscle Contraction By Reg. The Release Of Sequestered Calcium Ion
Resp. to Topologically Incorrect Protein
Gap Junction
Cilium Org.
Positive Reg. Endocytosis
MT Polymerization Or Depolymerization
Antimicrobial Mechanism Of IFN Stimulated Genes
DNA−Templated Transcription
WNT Signaling
Reg. Transmembrane Receptor Protein Serine Threonine Kinase Signaling
Signaling By Ntrks
Reg. Cytoplasmic Translation
Transport To The Golgi & Subsequent Mod.
Er To Golgi Anterograde Transport
Negative Reg. Cell Growth
Positive Reg. Establishment Of Protein Localization
Endocytic Recycling
Reg. DNA−Templated Transcription
MT Polymerization
Cellular Resp. to Heat Stress
Positive Reg. Post Translational Protein Mod.
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
GO_Slim
GOBP
KEGG_Legacy
Reactome
proteins
50
100
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_Interaction, fry sets at p < 0.05
107 sets. Page 3 of 3.
Cellular Resp. to Heat
Resp. to Heat
Transforming Growth Factor Beta Receptor Signaling
Mitotic Prometaphase
Positive Reg. WNT Signaling
Mitotic Spindle Checkpoint
Resp. to Temperature Stimulus
Cell Surface Receptor Protein Serine Threonine Kinase Signaling
Anchoring Of The Basal Body To The Plasma Membrane
Positive Reg. Protein Transport
Recruitment Of Mitotic Centrosome Proteins & Complexes
Aurka Activation By Tpx2
Reg. Plk1 Activity At G2 M Transition
Pathways In Cancer
Recruitment Of Numa To Mitotic Centrosomes
Renal Cell Carcinoma
Reg. Autophagy
Reg. Intracell. Transport
UPR
DNA Replication
mTOR Signalling
Reg. WNT Signaling
Reg. Canonical WNT Signaling
Resolution Of Sister Chromatid Cohesion
Autophagy
Resp. to Transforming Growth Factor Beta
0.0
0.5
1.0
1.5
2.0
signed −log10 p (positive: up)
proteins
30
60
90
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

# ANCOVA: HR minus LR at the later time point, at equal starting value

Training: 39 of 1378 sets at p < 0.05. Acute: 172 of 1378.




lm(later ~ earlier + arm) per set; training is T2 on T1, acute T3 on T2. Dashed line: p = 0.05. BH within window and collection. Table: c_data/02_contrasts.xlsx, ancova.